

P25 J3397: Non-uniform Refinement

CryoSPARC Running Version: v4.7.1

Project Details

DETAIL	VALUE
Title	24sep20b_WT-Control-White_From-January
Description	This is a repeat collection of White Control Wildtype CFTR from the original January/February 2024 collection. Using
ID	P25
Created by user	Andy Paige

Job Details

DETAIL	VALUE
Title	New Job J3397
Description	Enter a description.
ID	J3397
Job type	Non-uniform Refinement
Status	completed
Created by user	Minou Emmad
Created	Tue Jun 09 2026 15:13:18 GMT-0400 (Eastern Daylight Time)
Queued	Tue Jun 09 2026 15:13:54 GMT-0400 (Eastern Daylight Time)
Launched	Tue Jun 09 2026 15:13:58 GMT-0400 (Eastern Daylight Time)
Started	Tue Jun 09 2026 15:14:07 GMT-0400 (Eastern Daylight Time)
Waiting	-
Killed	-
Completed	Tue Jun 09 2026 15:34:46 GMT-0400 (Eastern Daylight Time)
Failed	-
Last accessed	Wed Jun 10 2026 15:54:19 GMT-0400 (Eastern Daylight Time)
Size	5.3 GB

Inputs

particles

J3392.particles_all_classes	
BLOB	
J3392.particles_all_classes.blob.F	
CTF	
J3392.particles_all_classes.ctf.F	
ALIGNMENTS3D	

J3392.particles_all_classes.alignments3D.F
PASSTHROUGH
J3392.particles_all_classes.alignments3D_multi.F

volume

J3392.volume_class_0
MAP
J3392.volume_class_0.map.F

mask

No group connections

Parameters

Particle preprocessing

PARAMETER	VALUE	DEFAULT	SPEC	ADVANCED
Window dataset (real-space)	True	X		
Window inner radius	0.85	X		X
Window outer radius	0.99	X		X

Homogeneous Refinement

PARAMETER	VALUE	DEFAULT	SPEC	ADVANCED
Symmetry	C1	X		
Symmetry relaxation method	none	X		X
Do symmetry alignment	True	X		X
Re-estimate greyscale level of input reference	True	X		X
Number of extra final passes	0	X		X
Maximum align resolution (A)	None	X		
Initial lowpass resolution (A)	30	X		X
GSFSC split resolution (A)	20	X		X
Force re-do GS split	False	X		X
Adaptive Marginalization	True	X		X
Non-uniform refine enable	True	X		X
Non-uniform filter order	8	X		X
Non-uniform AWF	3	X		X
Enforce non-negativity	False	X		X
Window structure in real space	True	X		X
Skip interpolant premult	True	X		X
Ignore DC component	True	X		X
Ignore tilt	False	X		X

Ignore trefoil	False	X		X	
Ignore tetra	False	X		X	
Use random ordering of particles	True	X		X	
Disable auto batchsize	False	X		X	
Batchsize epsilon	0.001	X		X	
Batchsize snrfactor	50	X		X	
Reset input per-particle scale	True	X		X	
Minimize over per-particle scale	False	X		X	
Scale min/use start iter	0	X		X	
Noise priorw	50	X		X	
Noise initw	200	X		X	
Noise initial sigma-scale	3	X		X	
Initialize noise model from images	False	X		X	
Dynamic mask threshold (0-1)	0.2	X		X	
Dynamic mask near (A)	6	X		X	
Dynamic mask far (A)	14	X		X	
Dynamic mask start resolution (A)	12	X			
Dynamic mask use absolute value	False	X		X	
Show plots from intermediate steps	True	X			
GPU batch size of images	None	X		X	

Defocus Refinement

PARAMETER	VALUE	DEFAULT	SPEC	ADVANCED	
Optimize per-particle defocus	False	X		X	
Num. particles to plot	3	X			
Minimum Fit Res (A)	20	X			
Defocus Search Range (A +/-)	2000	X			
GPU batch size of images	None	X		X	

Global CTF Refinement

PARAMETER	VALUE	DEFAULT	SPEC	ADVANCED	
Optimize per-group CTF params	False	X		X	
Num. groups to plot	3	X			
Binning to apply to plots	1	X		X	
Minimum Fit Res (A)	10	X			
Fit Tilt	True	X			
Fit Trefoil	True	X			

Fit Spherical Aberration	False	X			
Fit Tetrafoil	False	X			
Fit Anisotropic Mag.	False	X			
GPU batch size of images	None	X		X	

Ewald Sphere Correction

PARAMETER	VALUE	DEFAULT	SPEC	ADVANCED
Do EWS correction	False	X		X
EWS curvature sign	negative	X		X

Random Seeds

PARAMETER	VALUE	DEFAULT	SPEC	ADVANCED
Random seed	324059406	X		X

Compute settings

PARAMETER	VALUE	DEFAULT	SPEC	ADVANCED
Cache particle images on SSD	True	X		
Low-Memory Mode	False	X		

Outputs

particles

ALIGNMENTS3D	
J3397.particle.alignments3D	

CTF	
J3397.particle.ctf	

BLOB (PASSTHROUGH)	
J3397.particle.blob	

ALIGNMENTS3D_MULTI (PASSTHROUGH)	
J3397.particle.alignments3D_multi	

volume

MAP	
J3397.volume.blob	

MAP_SHARP	
J3397.volume.blob	

MAP_HALF_A	
------------	--

J3397.volume.blob	
MAP_HALF_B	
J3397.volume.blob	
MASK_REFINE	
J3397.volume.blob	
MASK_FSC	
J3397.volume.blob	
MASK_FSC_AUTO	
J3397.volume.blob	
PRECISION	
J3397.volume.blob	

mask

MASK_REFINE	
J3397.volume.blob	

Target

ganges.biology.columbia.edu (node)

```
> [2026-06-09 15:13:58.15 GMT-4] License is valid.
> [2026-06-09 15:13:58.15 GMT-4] Launching job on lane default target ganges.biology.columbia.edu ...
> [2026-06-09 15:13:58.18 GMT-4] Running job on master node hostname ganges.biology.columbia.edu
> [2026-06-09 15:14:07.78 GMT-4] [CPU: 78.9 MB]      [Avail: 337.57 GB] Job J3397 Started
> [2026-06-09 15:14:07.81 GMT-4] [CPU: 78.9 MB]      [Avail: 337.53 GB] Master running v4.7.1, worker running v4.7.1
> [2026-06-09 15:14:07.83 GMT-4] [CPU: 79.2 MB]      [Avail: 337.50 GB] Working in directory:
                                /mnt/nfs_HuntLabSynologyDS1621CryoEM/hCFTR/Movies/24sep20b/24
                                sep20b/rawdata/CS-24sep20b-wt-control-white-from-january/J339
                                7
> [2026-06-09 15:14:07.83 GMT-4] [CPU: 79.2 MB]      [Avail: 337.49 GB] Running on lane default
> [2026-06-09 15:14:07.83 GMT-4] [CPU: 79.2 MB]      [Avail: 337.49 GB] Resources allocated:
> [2026-06-09 15:14:07.84 GMT-4] [CPU: 79.2 MB]      [Avail: 337.48 GB]   Worker:  ganges.biology.columbia.edu
> [2026-06-09 15:14:07.84 GMT-4] [CPU: 79.2 MB]      [Avail: 337.47 GB]   CPU    :  [8, 9, 10, 11]
> [2026-06-09 15:14:07.84 GMT-4] [CPU: 79.2 MB]      [Avail: 337.47 GB]   GPU    :  [3]
> [2026-06-09 15:14:07.85 GMT-4] [CPU: 79.2 MB]      [Avail: 337.46 GB]   RAM    :  [4, 5, 6]
> [2026-06-09 15:14:07.85 GMT-4] [CPU: 79.2 MB]      [Avail: 337.46 GB]   SSD    :  True
> [2026-06-09 15:14:07.85 GMT-4] [CPU: 79.2 MB]      [Avail: 337.45 GB] -----
                                -
> [2026-06-09 15:14:07.86 GMT-4] [CPU: 79.2 MB]      [Avail: 337.44 GB] Importing job module for job type nonuniform_refine_new...
> [2026-06-09 15:14:19.06 GMT-4] [CPU: 273.7 MB]     [Avail: 336.78 GB] Job ready to run
> [2026-06-09 15:14:19.06 GMT-4] [CPU: 273.7 MB]     [Avail: 336.79 GB] *****
                                **
> [2026-06-09 15:14:26.85 GMT-4] [CPU: 302.2 MB]     [Avail: 337.35 GB] Using random seed of 324059406
> [2026-06-09 15:14:26.86 GMT-4] [CPU: 303.9 MB]     [Avail: 337.38 GB] Loading a ParticleStack with 27758 items...
> [2026-06-09 15:14:26.87 GMT-4] [CPU: 303.9 MB]     [Avail: 337.40 GB]

                                SSD cache ACTIVE at
                                /scr/instance_ganges.biology.columbia.edu:39001 (200 GB
                                reserve) (3 TB quota)

                                Cache usage                                Amount

                                Total / Usable                            3.58 TiB / 2.73 TiB
                                Used / Free                              2.73 TiB / 666.66 MiB
                                Hits / Misses                             81.91 GiB / 0.00 B
                                Acquired / Required                       81.91 GiB / 0.00 B

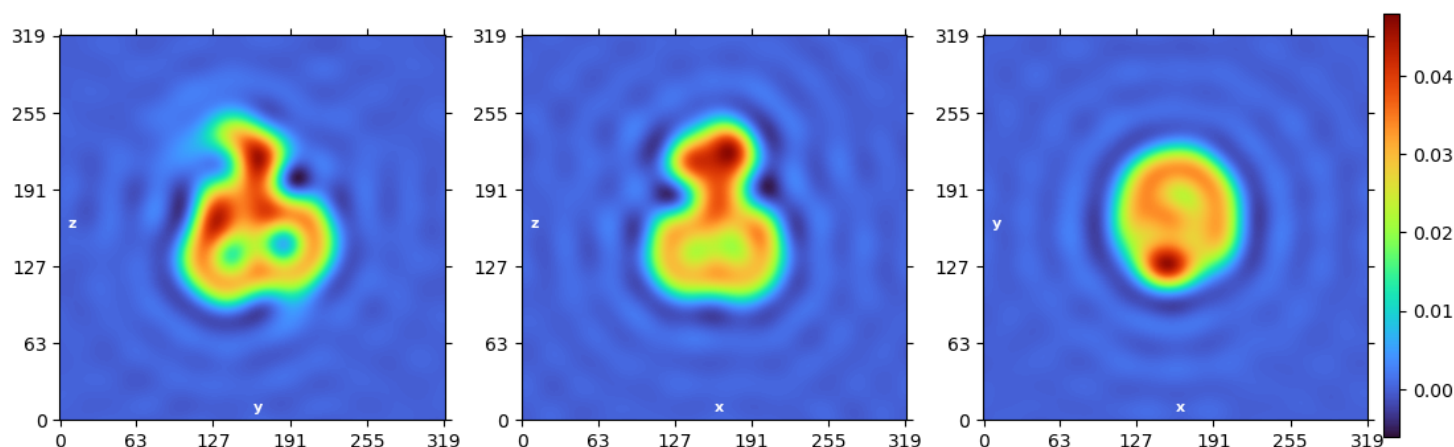
                                Progress:
                                [
                                8760/8760 (100%)
                                Elapsed: 0h 00m 07s
                                Active jobs: P25-J3396, P25-J3395, P25-J3397
                                SSD cache complete for 8760 file(s)

> [2026-06-09 15:14:36.03 GMT-4] [CPU: 358.8 MB]     [Avail: 336.98 GB] Done.
> [2026-06-09 15:14:36.03 GMT-4] [CPU: 358.8 MB]     [Avail: 336.98 GB] Windowing particles
> [2026-06-09 15:14:36.04 GMT-4] [CPU: 359.2 MB]     [Avail: 336.97 GB] Done.
> [2026-06-09 15:14:36.04 GMT-4] [CPU: 359.2 MB]     [Avail: 336.96 GB] ===== Gold Standard Split =====
> [2026-06-09 15:14:36.05 GMT-4] [CPU: 359.2 MB]     [Avail: 336.96 GB] Particles have input alignments3D connected, so reusing
                                pre-existing split
> [2026-06-09 15:14:36.05 GMT-4] [CPU: 359.2 MB]     [Avail: 336.95 GB] Set A is greater than set B by 102 particles (0.367 percent
                                difference relative to the total dataset).
> [2026-06-09 15:14:36.15 GMT-4] [CPU: 361.2 MB]     [Avail: 337.36 GB] Split A has 13930 particles
> [2026-06-09 15:14:36.16 GMT-4] [CPU: 361.2 MB]     [Avail: 337.36 GB] Split B has 13828 particles
> [2026-06-09 15:14:36.16 GMT-4] [CPU: 361.2 MB]     [Avail: 337.36 GB] ===== Refinement =====
> [2026-06-09 15:14:36.16 GMT-4] [CPU: 361.2 MB]     [Avail: 337.35 GB] Input particles have box size 320
> [2026-06-09 15:14:36.17 GMT-4] [CPU: 361.2 MB]     [Avail: 337.35 GB] Input particles have pixel size 0.8300
> [2026-06-09 15:14:36.17 GMT-4] [CPU: 361.2 MB]     [Avail: 337.34 GB] Particles will be zeropadded/truncated to size 320 during
                                alignment
```

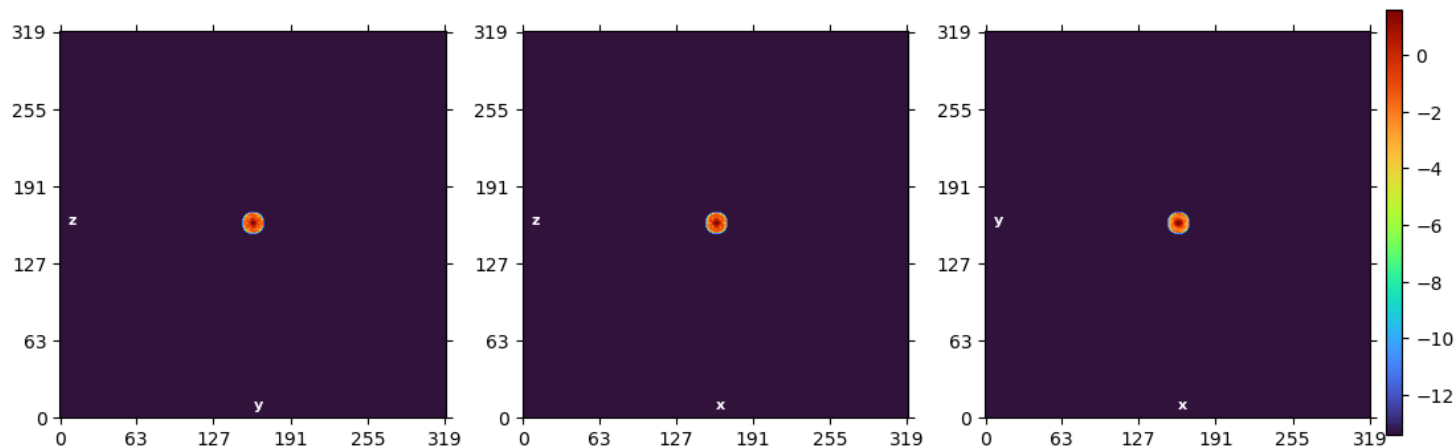
```

> [2026-06-09 15:14:36.17 GMT-4] [CPU: 361.5 MB] [Avail: 337.33 GB] Volume refinement will be done with effective box size 320
> [2026-06-09 15:14:36.17 GMT-4] [CPU: 361.5 MB] [Avail: 337.33 GB] Volume refinement will be done with pixel size 0.8300
> [2026-06-09 15:14:36.18 GMT-4] [CPU: 361.5 MB] [Avail: 337.32 GB] Particles will be zeropadded/truncated to size 320 during
backprojection
> [2026-06-09 15:14:36.18 GMT-4] [CPU: 361.5 MB] [Avail: 337.31 GB] Particles will be backprojected with box size 320
> [2026-06-09 15:14:36.18 GMT-4] [CPU: 361.5 MB] [Avail: 337.31 GB] Volume will be internally cropped and stored with box size
320
> [2026-06-09 15:14:36.19 GMT-4] [CPU: 361.5 MB] [Avail: 337.30 GB] Volume will be interpolated with box size 320 (zeropadding
factor 1.00)
> [2026-06-09 15:14:36.19 GMT-4] [CPU: 361.5 MB] [Avail: 337.29 GB] DC components of images will be ignored and volume will be
floated at each iteration.
> [2026-06-09 15:14:36.19 GMT-4] [CPU: 361.5 MB] [Avail: 337.29 GB] Spherical windowing of maps is enabled
> [2026-06-09 15:14:36.19 GMT-4] [CPU: 361.5 MB] [Avail: 337.29 GB] Refining with C1 symmetry enforced
> [2026-06-09 15:14:36.20 GMT-4] [CPU: 361.5 MB] [Avail: 337.29 GB] Refining with pose and shift marginalization during
backprojection enabled.
> [2026-06-09 15:14:36.20 GMT-4] [CPU: 361.5 MB] [Avail: 337.29 GB] Resetting input per-particle scale factors to 1.0
> [2026-06-09 15:14:36.20 GMT-4] [CPU: 361.5 MB] [Avail: 337.29 GB] Starting at initial resolution 30.000A (radwn 8.853).
> [2026-06-09 15:14:36.21 GMT-4] [CPU: 361.5 MB] [Avail: 337.29 GB] ===== Non-Uniform Refinement =====
> [2026-06-09 15:14:36.21 GMT-4] [CPU: 361.5 MB] [Avail: 337.29 GB] Non-Uniform Refinement is enabled.
> [2026-06-09 15:14:36.21 GMT-4] [CPU: 361.5 MB] [Avail: 337.28 GB] Using AWF of 3.00.
> [2026-06-09 15:14:36.21 GMT-4] [CPU: 361.5 MB] [Avail: 337.28 GB] Using butterworth filter with order 8.
> [2026-06-09 15:14:36.22 GMT-4] [CPU: 361.7 MB] [Avail: 337.28 GB] ===== Masking =====
> [2026-06-09 15:14:38.53 GMT-4] [CPU: 894.1 MB] [Avail: 335.71 GB] No mask input was connected, so dynamic masking will be
enabled.
> [2026-06-09 15:14:38.53 GMT-4] [CPU: 894.2 MB] [Avail: 335.71 GB] Dynamic mask threshold: 0.2000
> [2026-06-09 15:14:38.53 GMT-4] [CPU: 894.2 MB] [Avail: 335.71 GB] Dynamic mask near (A): 6.00
> [2026-06-09 15:14:38.54 GMT-4] [CPU: 894.2 MB] [Avail: 335.71 GB] Dynamic mask far (A): 14.00
> [2026-06-09 15:14:38.54 GMT-4] [CPU: 894.2 MB] [Avail: 335.71 GB] ===== Initial Model =====
> [2026-06-09 15:14:38.54 GMT-4] [CPU: 894.2 MB] [Avail: 335.70 GB] Resampling initial model to specified volume representation
size and pixel-size...
> [2026-06-09 15:14:41.72 GMT-4] [CPU: 1.29 GB] [Avail: 335.67 GB] Estimating scale of initial reference.
> [2026-06-09 15:14:47.12 GMT-4] [CPU: 77.4 MB] [Avail: 333.05 GB] WARNING: io_uring support disabled (not supported by kernel),
I/O performance may degrade
> [2026-06-09 15:14:48.09 GMT-4] [CPU: 1.49 GB] [Avail: 334.90 GB] Rescaling initial reference by a factor of 1.440
> [2026-06-09 15:14:49.72 GMT-4] [CPU: 1.49 GB] [Avail: 334.54 GB] Estimating scale of initial reference.
> [2026-06-09 15:14:52.57 GMT-4] [CPU: 1.45 GB] [Avail: 335.59 GB] Rescaling initial reference by a factor of 1.000
> [2026-06-09 15:14:54.15 GMT-4] [CPU: 1.45 GB] [Avail: 335.99 GB] Estimating scale of initial reference.
> [2026-06-09 15:14:57.04 GMT-4] [CPU: 1.45 GB] [Avail: 336.41 GB] Rescaling initial reference by a factor of 1.002
> [2026-06-09 15:15:00.40 GMT-4] Initial Real Space Slices

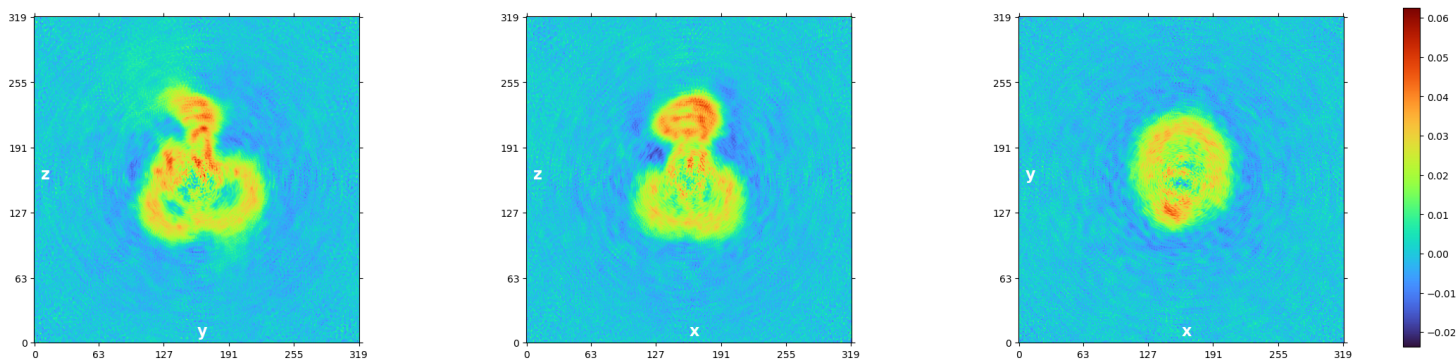
```



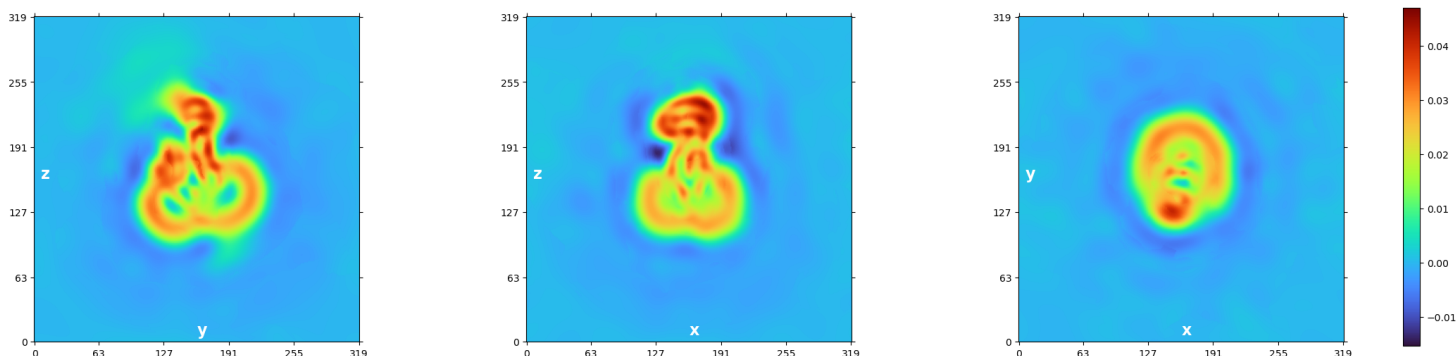
> [2026-06-09 15:15:01.68 GMT-4] Initial Fourier Space Slices



```
> [2026-06-09 15:15:01.69 GMT-4] [CPU: 1.50 GB] [Avail: 335.44 GB] ===== Starting Refinement Iterations =====
> [2026-06-09 15:15:01.69 GMT-4] [CPU: 1.50 GB] [Avail: 335.43 GB] ----- Start Iteration 0
> [2026-06-09 15:15:01.69 GMT-4] [CPU: 1.50 GB] [Avail: 335.43 GB] Using Max Alignment Radius 8.853 (30.000A)
> [2026-06-09 15:15:01.70 GMT-4] [CPU: 1.50 GB] [Avail: 335.43 GB] Auto batchsize: 5942 in each split
> [2026-06-09 15:15:05.20 GMT-4] [CPU: 2.01 GB] [Avail: 333.50 GB] -- THR 1 BATCH 500 NUM 1471 TOTAL 3.1482815 ELAPSED 30.181350
--
> [2026-06-09 15:15:37.65 GMT-4] [CPU: 3.21 GB] [Avail: 327.69 GB] Processed 11884.000 images in 32.446s.
> [2026-06-09 15:15:41.83 GMT-4] [CPU: 3.53 GB] [Avail: 327.45 GB] Computing FSCs...
> [2026-06-09 15:15:41.84 GMT-4] [CPU: 3.53 GB] [Avail: 327.44 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-09 15:15:41.84 GMT-4] [CPU: 3.53 GB] [Avail: 327.44 GB] Computing FFTs on GPU.
> [2026-06-09 15:15:44.45 GMT-4] [CPU: 3.78 GB] [Avail: 326.98 GB] Done in 2.610s
> [2026-06-09 15:15:44.45 GMT-4] [CPU: 3.78 GB] [Avail: 326.97 GB] Computing cFSCs...
> [2026-06-09 15:15:46.45 GMT-4] [CPU: 3.78 GB] [Avail: 327.12 GB] Done in 2.000s
> [2026-06-09 15:15:46.46 GMT-4] [CPU: 3.78 GB] [Avail: 327.11 GB] Using Filter Radius 32.165 (8.257A) | Previous: 8.853
(30.000A)
> [2026-06-09 15:15:52.42 GMT-4] [CPU: 4.85 GB] [Avail: 325.67 GB] Non-uniform regularization with compute option: GPU
> [2026-06-09 15:15:52.42 GMT-4] [CPU: 4.85 GB] [Avail: 325.66 GB] Running local cross validation for A ...
> [2026-06-09 15:16:20.90 GMT-4] [CPU: 7.52 GB] [Avail: 321.78 GB] Local cross validation A done in 28.474s
> [2026-06-09 15:16:22.51 GMT-4] FSC Filtered Side A
```

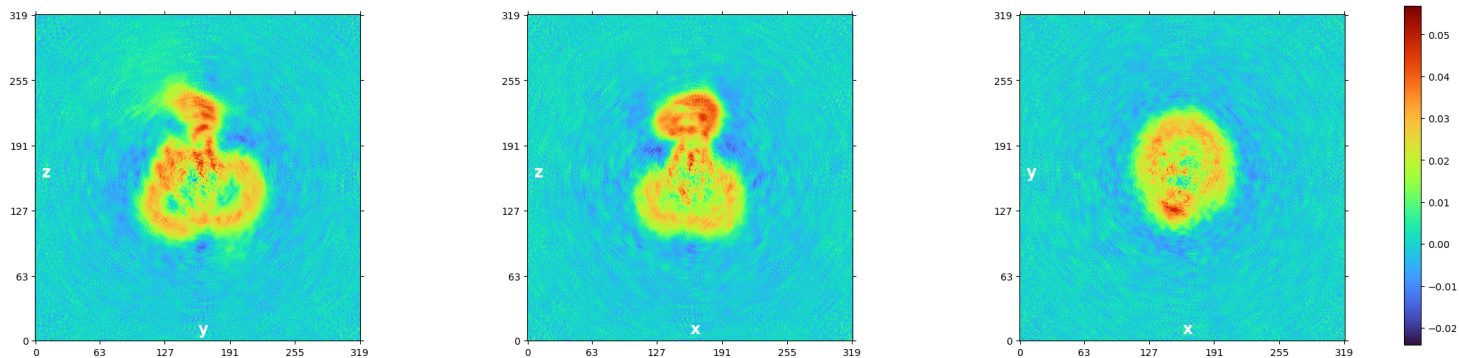


> [2026-06-09 15:16:23.65 GMT-4] CV Filtered Side A

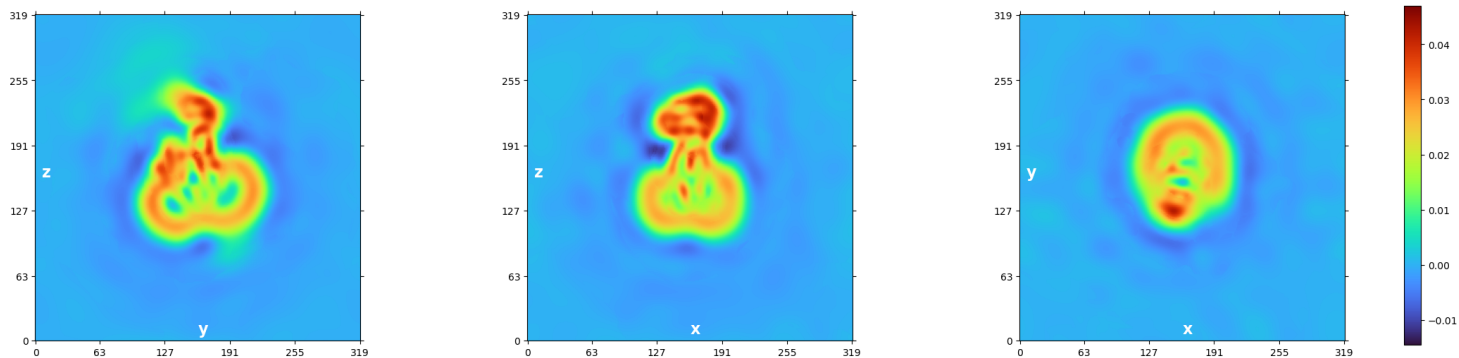


```
> [2026-06-09 15:16:23.65 GMT-4] [CPU: 7.53 GB] [Avail: 322.32 GB] Running local cross validation for B ...
> [2026-06-09 15:16:45.69 GMT-4] [CPU: 7.81 GB] [Avail: 317.99 GB] Local cross validation B done in 22.029s
```


> [2026-06-09 15:16:47.20 GMT-4] FSC Filtered Side B



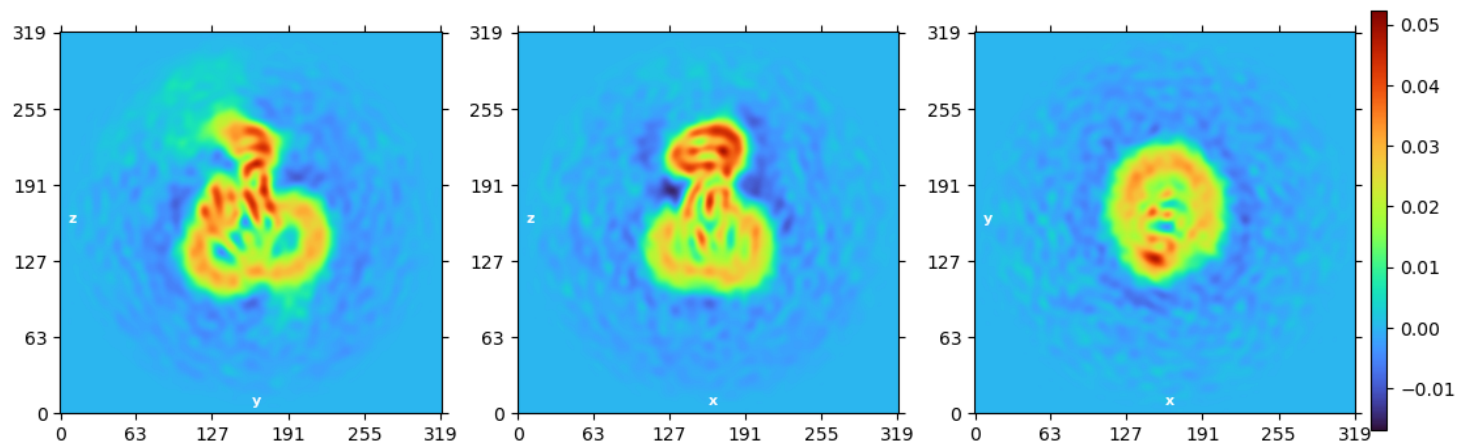
> [2026-06-09 15:16:48.31 GMT-4] CV Filtered Side B



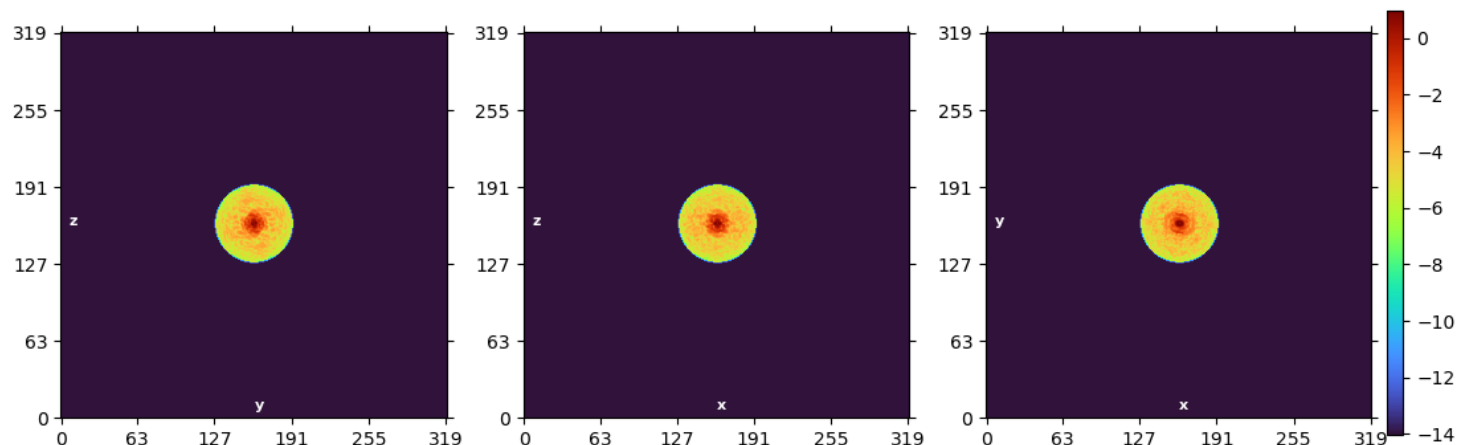
> [2026-06-09 15:16:56.78 GMT-4] [CPU: 8.14 GB] [Avail: 317.61 GB] Estimated Bfactor: -349.4

> [2026-06-09 15:16:56.78 GMT-4] [CPU: 8.14 GB] [Avail: 317.60 GB] Plotting..

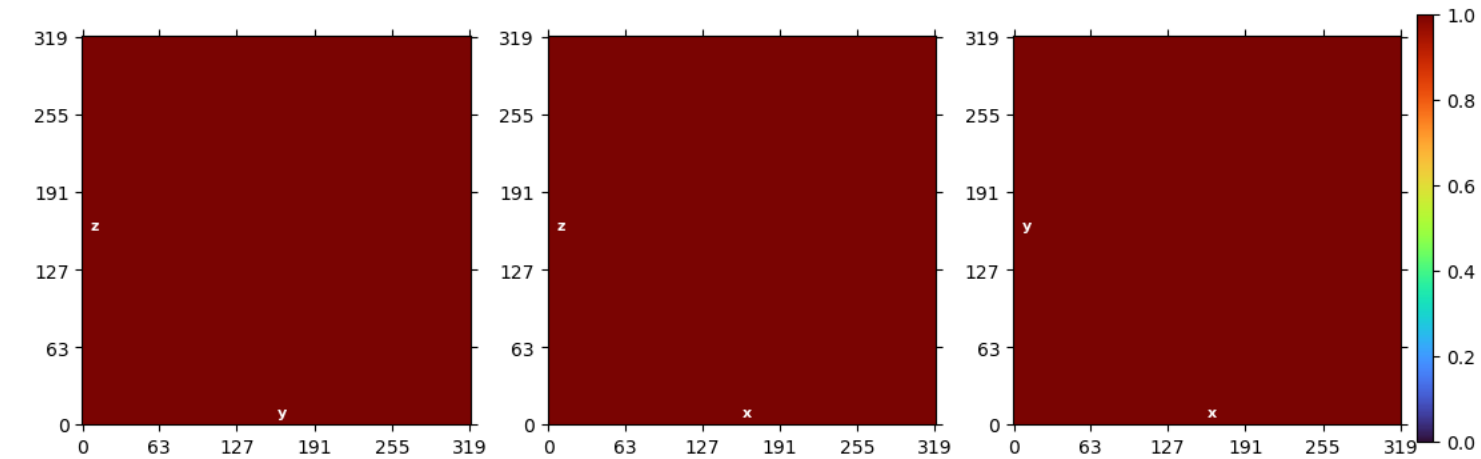
> [2026-06-09 15:17:03.45 GMT-4] Real Space Slices Iteration 000



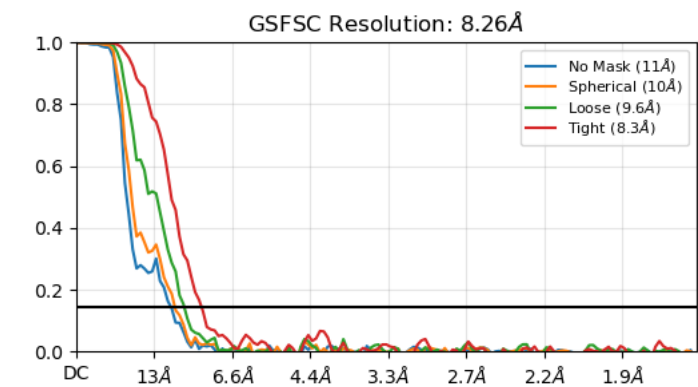
> [2026-06-09 15:17:04.84 GMT-4] Fourier Space Slices Iteration 000



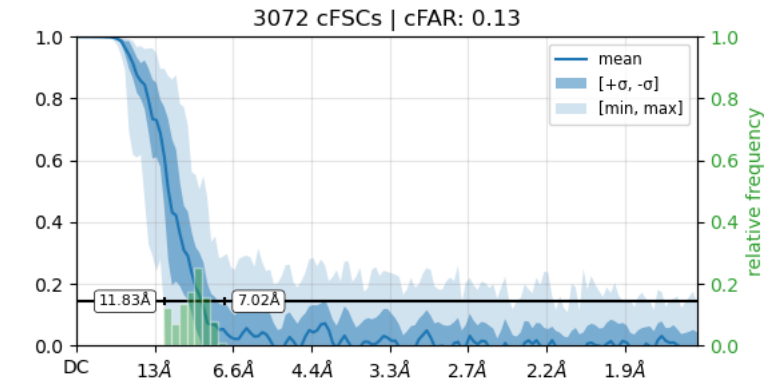
> [2026-06-09 15:17:05.97 GMT-4] Real Space Mask Slices Iteration 000



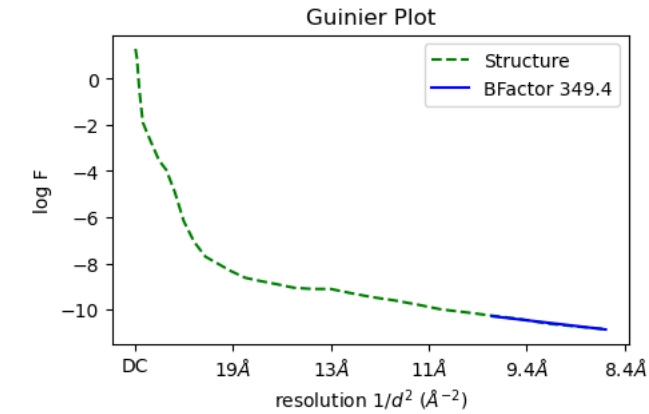
> [2026-06-09 15:17:06.56 GMT-4] FSC Iteration 000



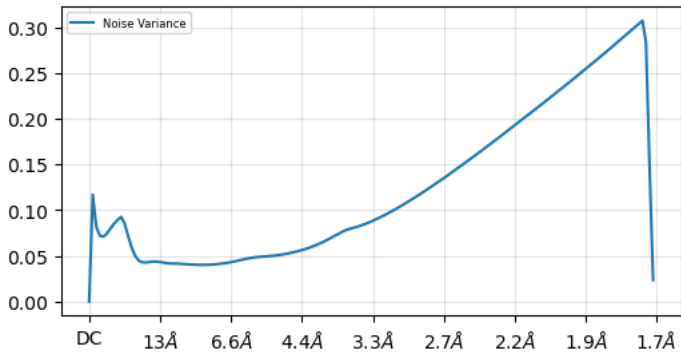
> [2026-06-09 15:17:07.31 GMT-4] cFSCs (Half-angle: 20°) Iteration 000, with tight mask



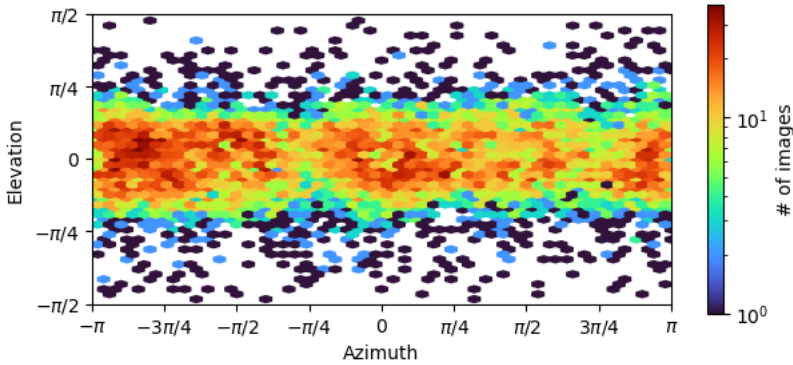
> [2026-06-09 15:17:12.73 GMT-4] Guinier Plot Iteration 000



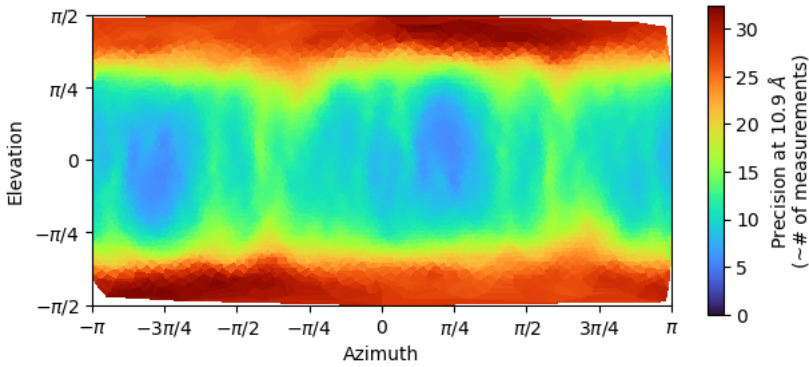
> [2026-06-09 15:17:13.12 GMT-4] Noise Model Iteration 000



> [2026-06-09 15:17:14.58 GMT-4] Viewing Direction Distribution Iteration 000

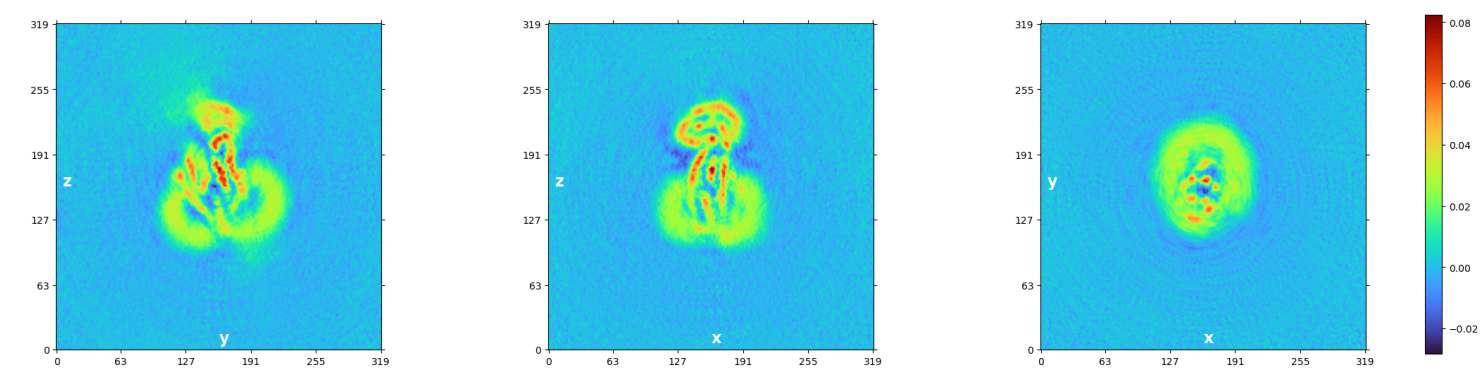


> [2026-06-09 15:17:16.39 GMT-4] Posterior Precision Directional Distribution Iteration 000

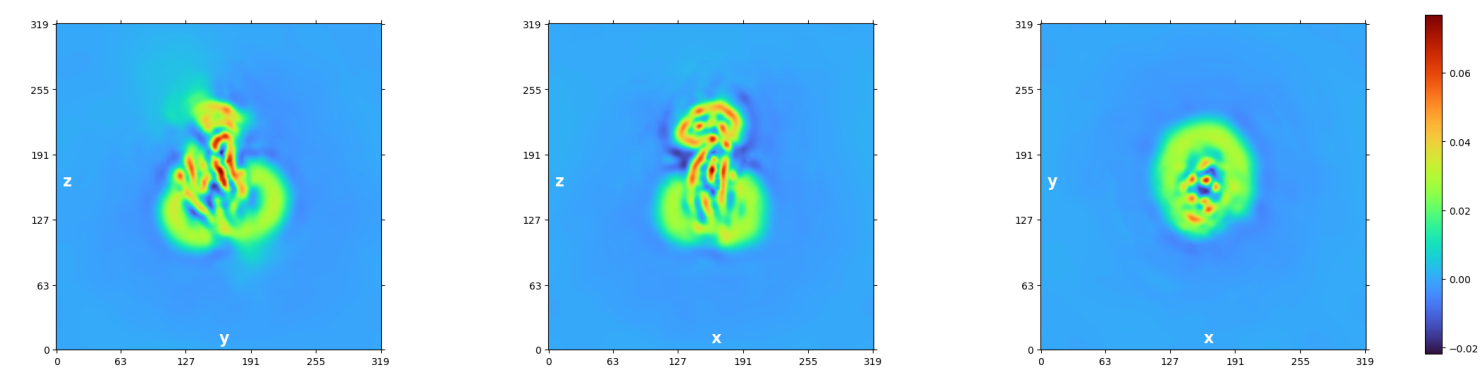


```
> [2026-06-09 15:17:16.39 GMT-4] [CPU: 8.56 GB] [Avail: 317.10 GB] Done in 19.602s.
> [2026-06-09 15:17:16.39 GMT-4] [CPU: 8.56 GB] [Avail: 317.10 GB] Outputting files..
> [2026-06-09 15:17:32.30 GMT-4] [CPU: 9.00 GB] [Avail: 315.81 GB] Done in 15.899s.
> [2026-06-09 15:17:32.30 GMT-4] [CPU: 9.00 GB] [Avail: 315.81 GB] Done iteration 0 in 150.607s. Total time so far 185.791s
> [2026-06-09 15:17:33.02 GMT-4] [CPU: 9.00 GB] [Avail: 315.43 GB] ----- Start Iteration 1
> [2026-06-09 15:17:33.02 GMT-4] [CPU: 9.00 GB] [Avail: 315.43 GB] Using Max Alignment Radius 32.165 (8.257A)
> [2026-06-09 15:17:33.02 GMT-4] [CPU: 9.00 GB] [Avail: 315.43 GB] Auto batchsize: 13930 in each split
> [2026-06-09 15:17:33.03 GMT-4] [CPU: 9.00 GB] [Avail: 315.43 GB] Using dynamic mask.
> [2026-06-09 15:18:03.74 GMT-4] [CPU: 9.50 GB] [Avail: 312.44 GB] -- THR 0 BATCH 500 NUM 3414 TOTAL 6.2005884 ELAPSED 54.255921
--
> [2026-06-09 15:19:00.53 GMT-4] [CPU: 10.36 GB] [Avail: 295.39 GB] Processed 27758.000 images in 56.789s.
> [2026-06-09 15:19:04.08 GMT-4] [CPU: 10.39 GB] [Avail: 297.60 GB] Computing FSCs...
> [2026-06-09 15:19:04.08 GMT-4] [CPU: 10.39 GB] [Avail: 297.60 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-09 15:19:04.09 GMT-4] [CPU: 10.39 GB] [Avail: 297.60 GB] Computing FFTs on GPU.
> [2026-06-09 15:19:06.23 GMT-4] [CPU: 10.39 GB] [Avail: 296.37 GB] Done in 2.143s
> [2026-06-09 15:19:06.23 GMT-4] [CPU: 10.39 GB] [Avail: 296.37 GB] Computing cFSCs...
> [2026-06-09 15:19:08.40 GMT-4] [CPU: 10.39 GB] [Avail: 296.28 GB] Done in 2.164s
> [2026-06-09 15:19:08.40 GMT-4] [CPU: 10.39 GB] [Avail: 296.28 GB] Using Filter Radius 57.702 (4.603A) | Previous: 32.165
(8.257A)
> [2026-06-09 15:19:15.54 GMT-4] [CPU: 9.64 GB] [Avail: 298.58 GB] Non-uniform regularization with compute option: GPU
> [2026-06-09 15:19:15.54 GMT-4] [CPU: 9.64 GB] [Avail: 298.55 GB] Running local cross validation for A ...
> [2026-06-09 15:19:40.27 GMT-4] [CPU: 9.86 GB] [Avail: 297.83 GB] Local cross validation A done in 24.721s
```


> [2026-06-09 15:19:41.85 GMT-4] FSC Filtered Side A



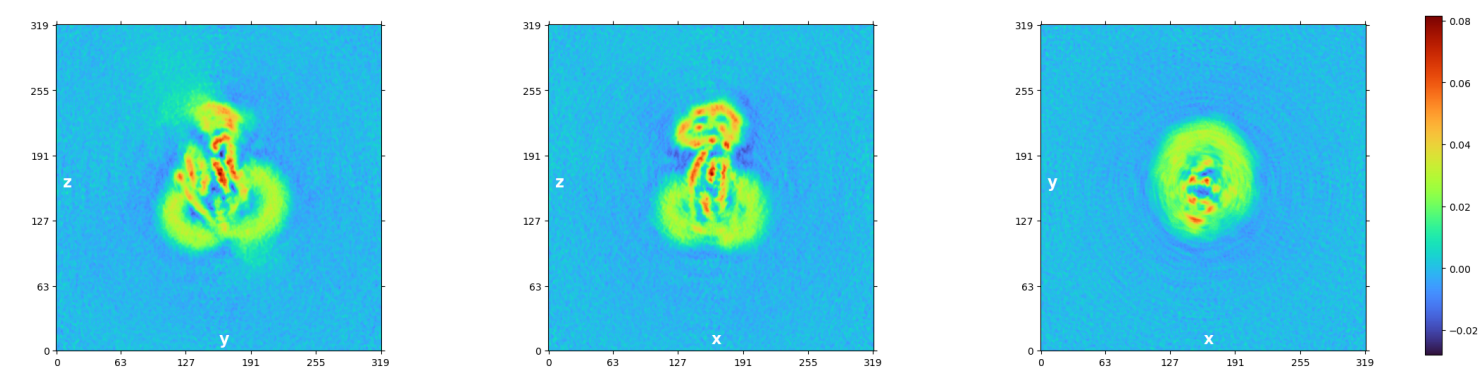
> [2026-06-09 15:19:43.08 GMT-4] CV Filtered Side A



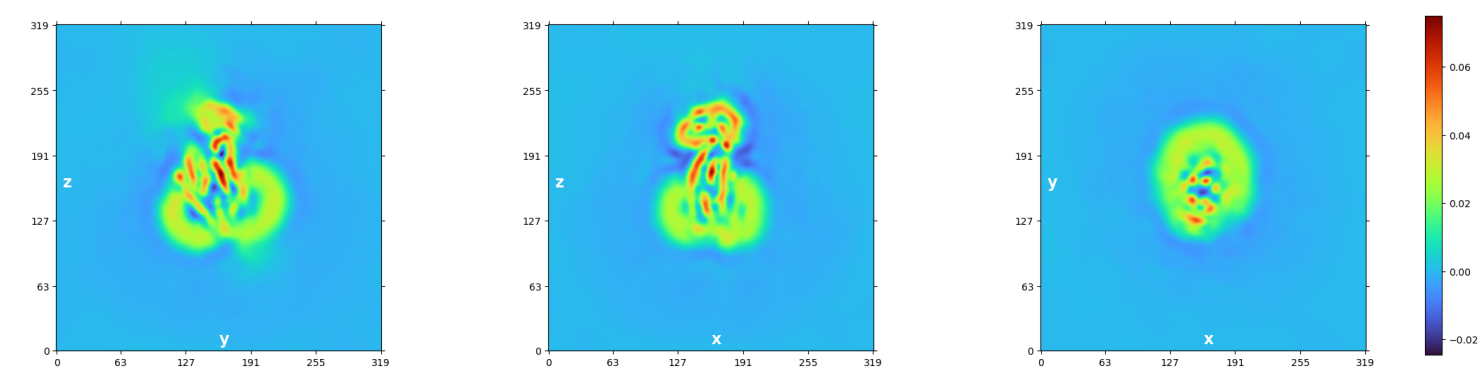
> [2026-06-09 15:19:43.08 GMT-4] [CPU: 9.86 GB] [Avail: 298.71 GB] Running local cross validation for B ...

> [2026-06-09 15:20:06.04 GMT-4] [CPU: 10.11 GB] [Avail: 298.98 GB] Local cross validation B done in 22.958s

> [2026-06-09 15:20:07.60 GMT-4] FSC Filtered Side B



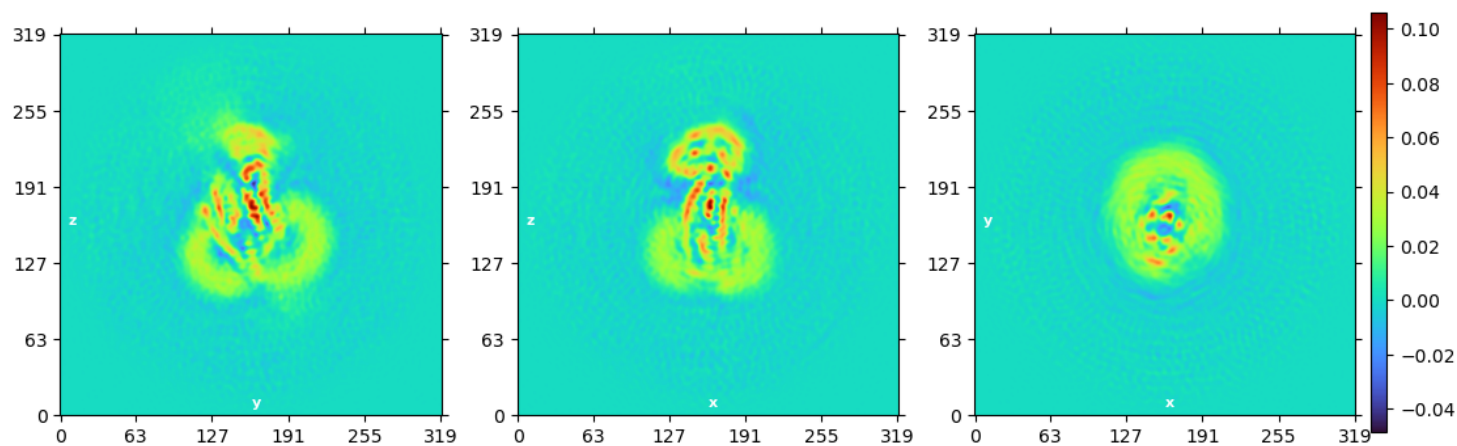
> [2026-06-09 15:20:08.66 GMT-4] CV Filtered Side B



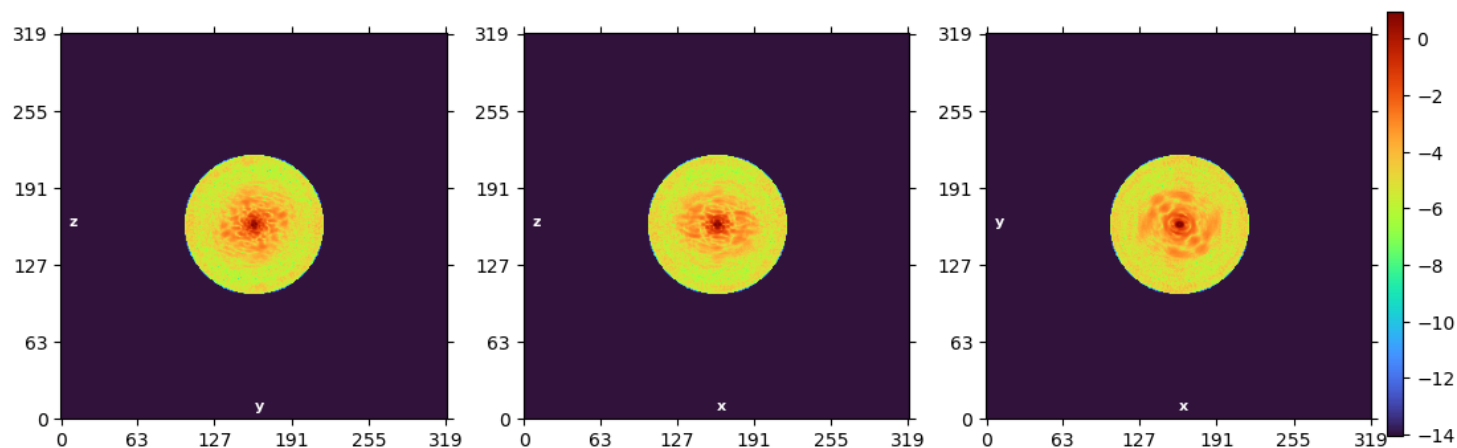
> [2026-06-09 15:20:19.32 GMT-4] [CPU: 10.35 GB] [Avail: 298.39 GB] Estimated Bfactor: -184.8

> [2026-06-09 15:20:19.33 GMT-4] [CPU: 10.35 GB] [Avail: 298.39 GB] Plotting..

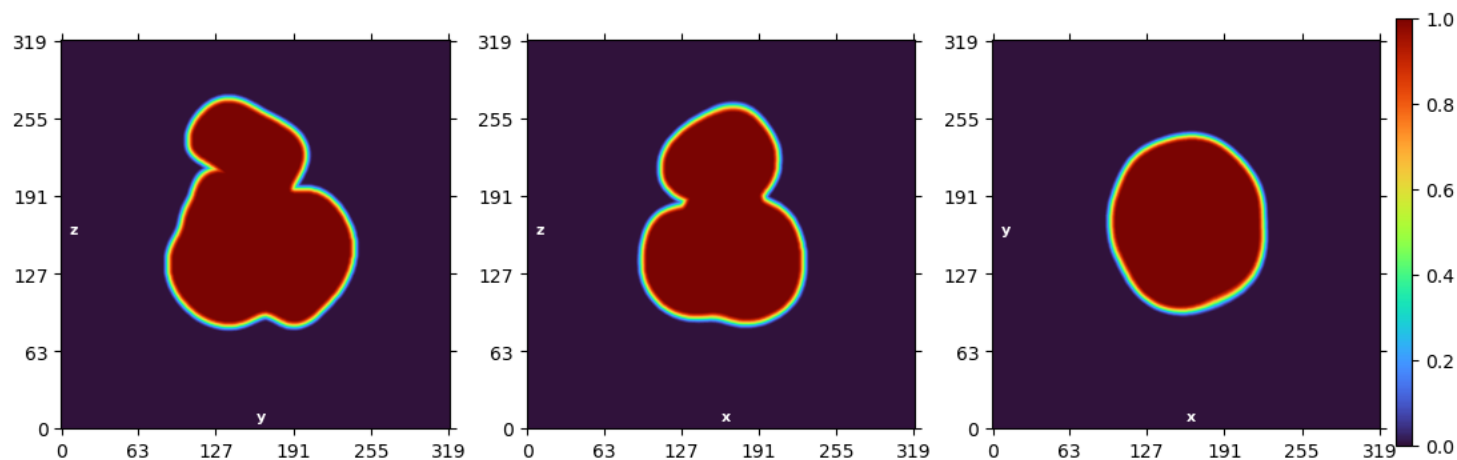
> [2026-06-09 15:20:25.88 GMT-4] Real Space Slices Iteration 001



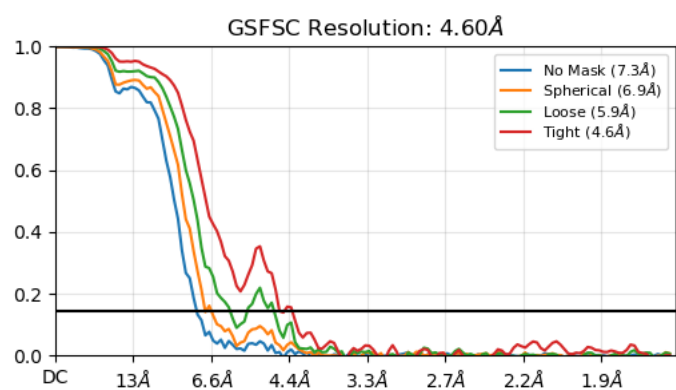
> [2026-06-09 15:20:27.18 GMT-4] Fourier Space Slices Iteration 001



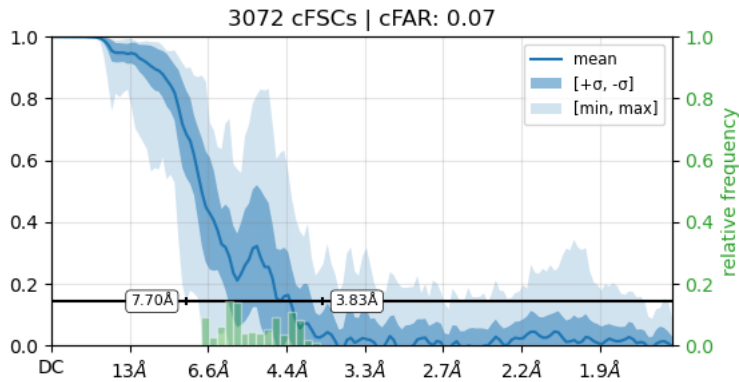
> [2026-06-09 15:20:29.19 GMT-4] Real Space Mask Slices Iteration 001



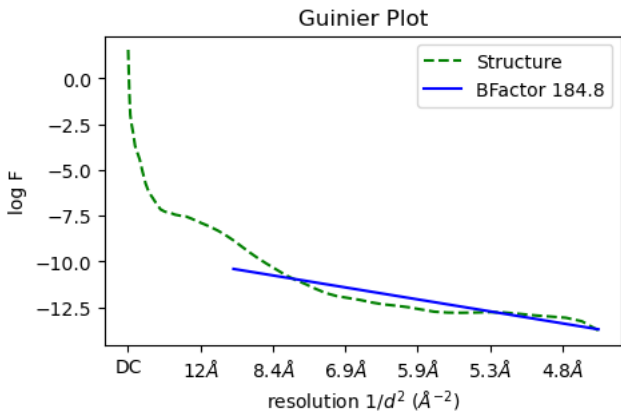
> [2026-06-09 15:20:29.73 GMT-4] FSC Iteration 001



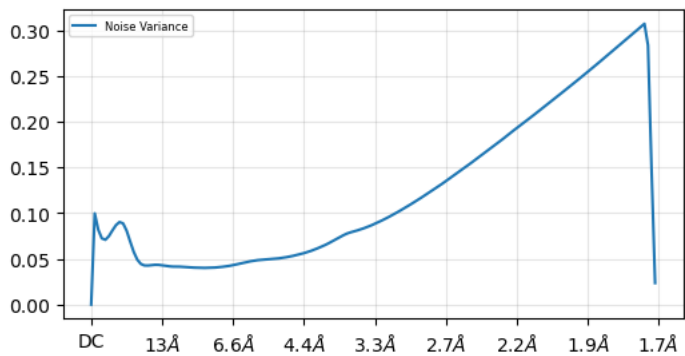
> [2026-06-09 15:20:30.54 GMT-4] cFSCs (Half-angle: 20°) Iteration 001, with tight mask



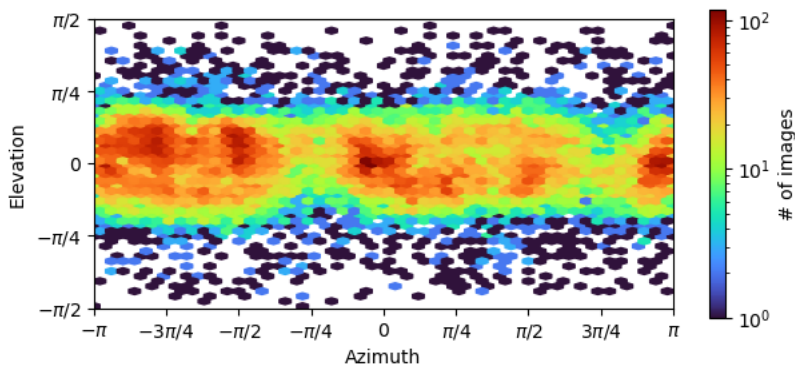
> [2026-06-09 15:20:36.28 GMT-4] Guinier Plot Iteration 001



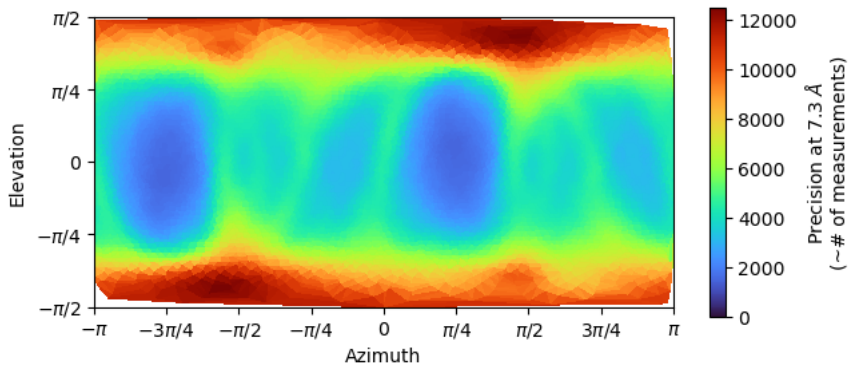
> [2026-06-09 15:20:36.66 GMT-4] Noise Model Iteration 001



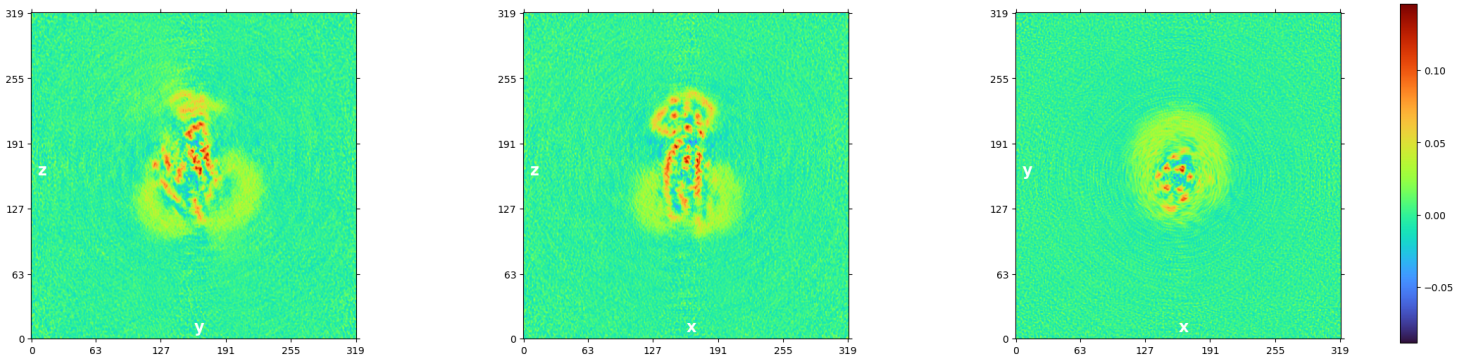
> [2026-06-09 15:20:37.65 GMT-4] Viewing Direction Distribution Iteration 001



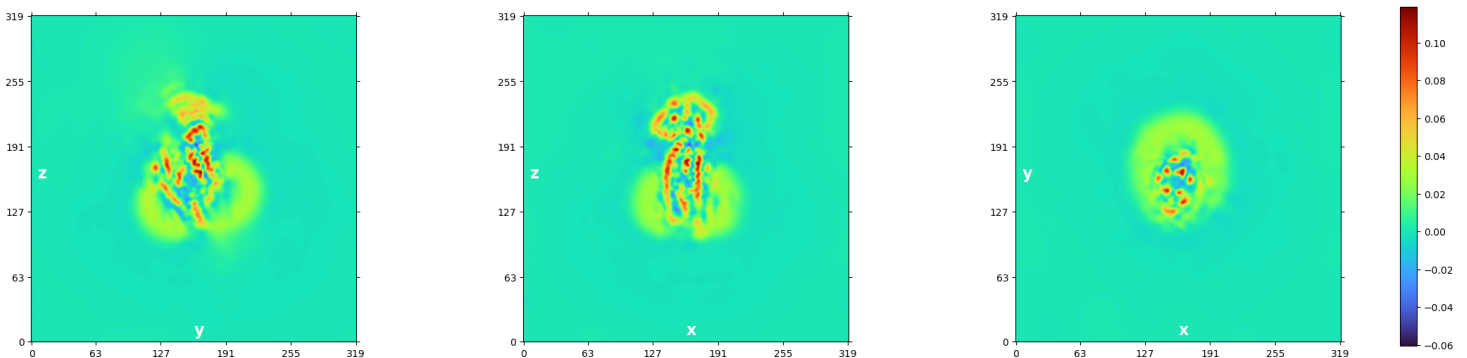
> [2026-06-09 15:20:39.26 GMT-4] Posterior Precision Directional Distribution Iteration 001



```
> [2026-06-09 15:20:39.26 GMT-4] [CPU: 10.39 GB] [Avail: 297.63 GB] Done in 19.932s.
> [2026-06-09 15:20:39.27 GMT-4] [CPU: 10.39 GB] [Avail: 297.62 GB] Outputting files..
> [2026-06-09 15:20:55.10 GMT-4] [CPU: 10.30 GB] [Avail: 295.46 GB] Done in 15.828s.
> [2026-06-09 15:20:55.10 GMT-4] [CPU: 10.30 GB] [Avail: 295.49 GB] Done iteration 1 in 202.080s. Total time so far 388.591s
> [2026-06-09 15:20:56.06 GMT-4] [CPU: 10.30 GB] [Avail: 293.90 GB] ----- Start Iteration 2
> [2026-06-09 15:20:56.07 GMT-4] [CPU: 10.30 GB] [Avail: 293.82 GB] Using Max Alignment Radius 57.702 (4.603A)
> [2026-06-09 15:20:56.07 GMT-4] [CPU: 10.30 GB] [Avail: 293.81 GB] Auto batchsize: 13930 in each split
> [2026-06-09 15:20:56.07 GMT-4] [CPU: 10.30 GB] [Avail: 293.81 GB] Using dynamic mask.
> [2026-06-09 15:21:27.99 GMT-4] [CPU: 10.30 GB] [Avail: 292.31 GB] -- THR 1 BATCH 500 NUM 3414 TOTAL 7.0909605 ELAPSED 56.216471
--
> [2026-06-09 15:22:26.26 GMT-4] [CPU: 11.15 GB] [Avail: 294.70 GB] Processed 27758.000 images in 58.270s.
> [2026-06-09 15:22:29.43 GMT-4] [CPU: 11.15 GB] [Avail: 294.30 GB] Computing FSCs...
> [2026-06-09 15:22:29.44 GMT-4] [CPU: 11.15 GB] [Avail: 294.29 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-09 15:22:29.44 GMT-4] [CPU: 11.15 GB] [Avail: 294.29 GB] Computing FFTs on GPU.
> [2026-06-09 15:22:31.25 GMT-4] [CPU: 11.15 GB] [Avail: 292.42 GB] Done in 1.811s
> [2026-06-09 15:22:31.25 GMT-4] [CPU: 11.15 GB] [Avail: 292.41 GB] Computing cFSCs...
> [2026-06-09 15:22:33.31 GMT-4] [CPU: 11.15 GB] [Avail: 292.83 GB] Done in 2.053s
> [2026-06-09 15:22:33.31 GMT-4] [CPU: 11.15 GB] [Avail: 292.83 GB] Using Filter Radius 72.190 (3.679A) | Previous: 57.702
(4.603A)
> [2026-06-09 15:22:39.57 GMT-4] [CPU: 10.40 GB] [Avail: 292.31 GB] Non-uniform regularization with compute option: GPU
> [2026-06-09 15:22:39.57 GMT-4] [CPU: 10.40 GB] [Avail: 292.31 GB] Running local cross validation for A ...
> [2026-06-09 15:23:03.07 GMT-4] [CPU: 10.65 GB] [Avail: 293.60 GB] Local cross validation A done in 23.498s
> [2026-06-09 15:23:04.47 GMT-4] FSC Filtered Side A
```



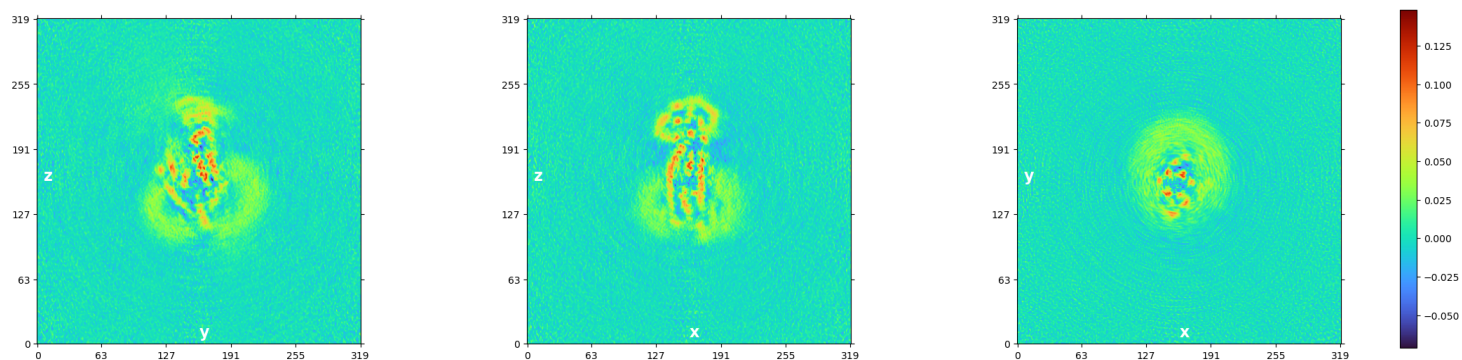
> [2026-06-09 15:23:05.50 GMT-4] CV Filtered Side A



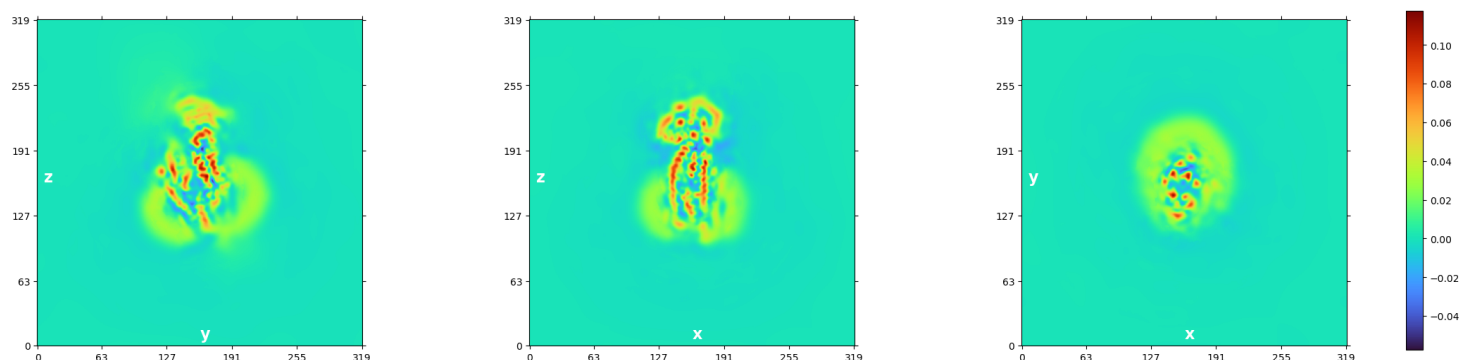
> [2026-06-09 15:23:05.50 GMT-4] [CPU: 10.65 GB] [Avail: 294.94 GB] Running local cross validation for B ...

> [2026-06-09 15:23:27.76 GMT-4] [CPU: 10.90 GB] [Avail: 292.69 GB] Local cross validation B done in 22.257s

> [2026-06-09 15:23:29.27 GMT-4] FSC Filtered Side B



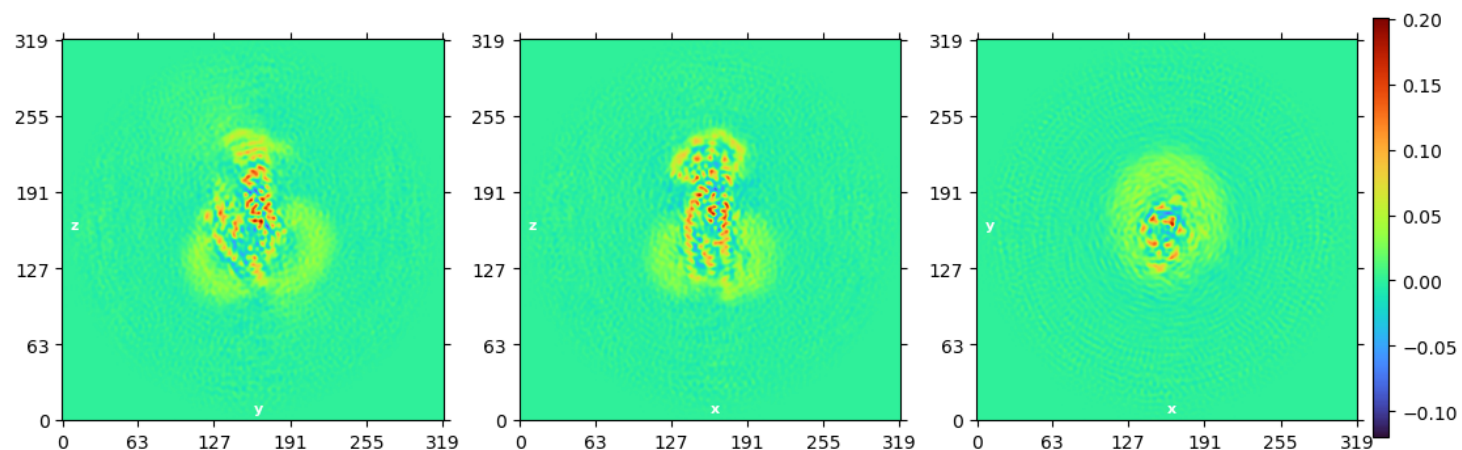
> [2026-06-09 15:23:30.31 GMT-4] CV Filtered Side B



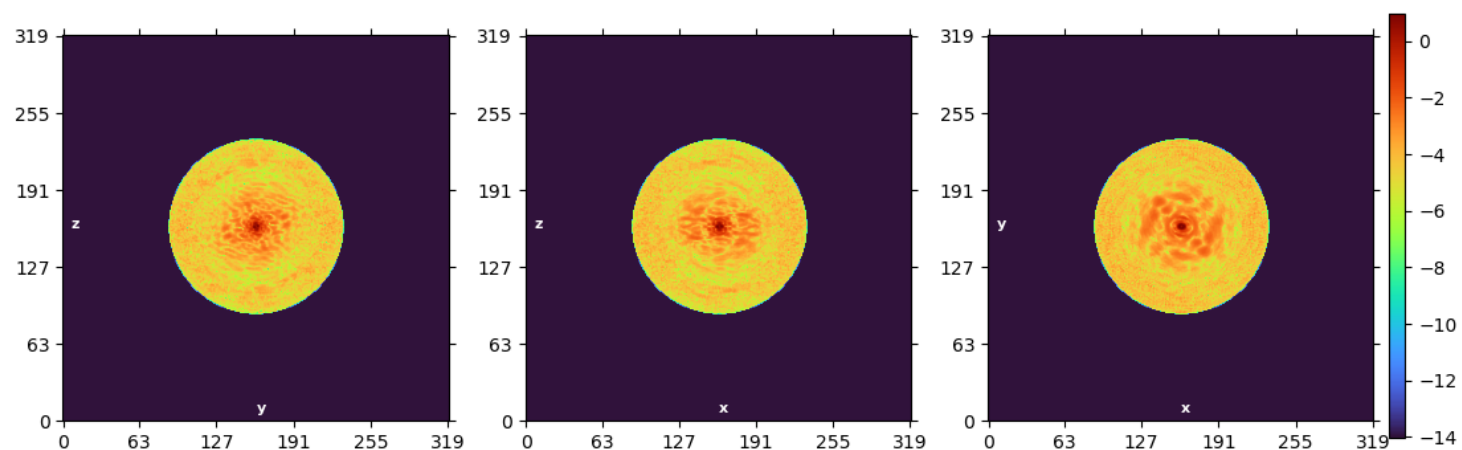
> [2026-06-09 15:23:41.19 GMT-4] [CPU: 10.93 GB] [Avail: 295.25 GB] Estimated Bfactor: -105.7

> [2026-06-09 15:23:41.20 GMT-4] [CPU: 10.93 GB] [Avail: 295.24 GB] Plotting..

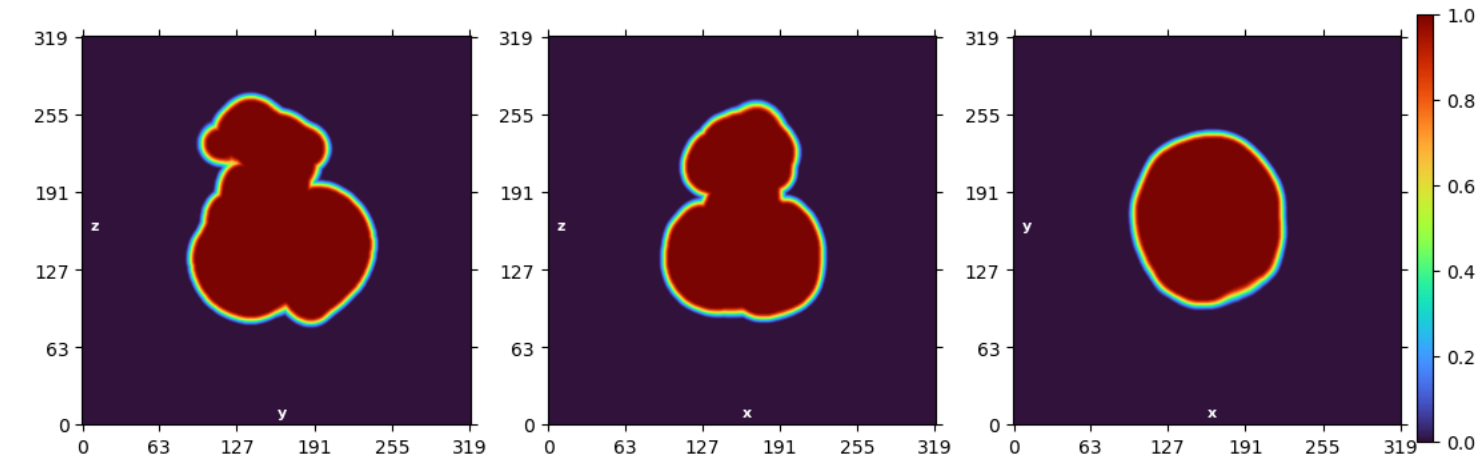
> [2026-06-09 15:23:47.45 GMT-4] Real Space Slices Iteration 002



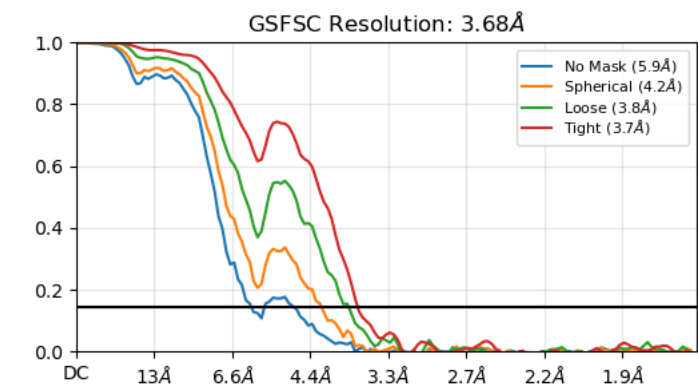
> [2026-06-09 15:23:48.82 GMT-4] Fourier Space Slices Iteration 002



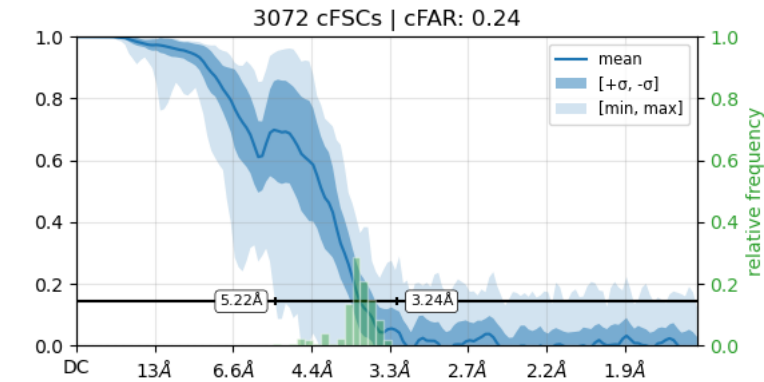
> [2026-06-09 15:23:50.64 GMT-4] Real Space Mask Slices Iteration 002



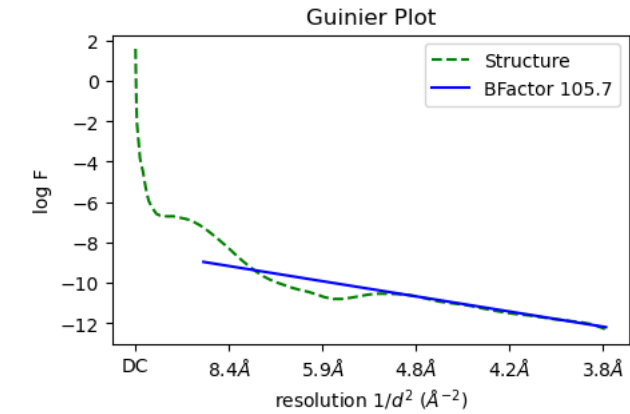
> [2026-06-09 15:23:51.20 GMT-4] FSC Iteration 002



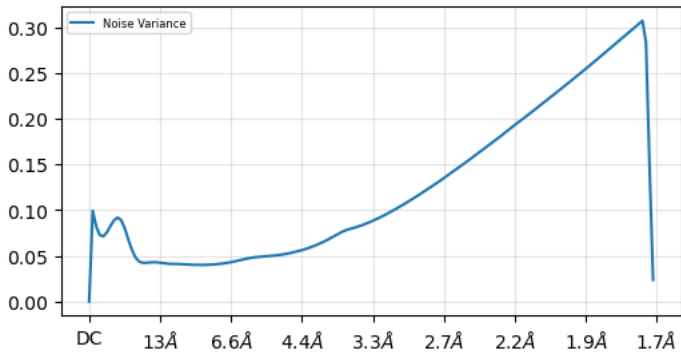
> [2026-06-09 15:23:52.05 GMT-4] cFSCs (Half-angle: 20°) Iteration 002, with tight mask



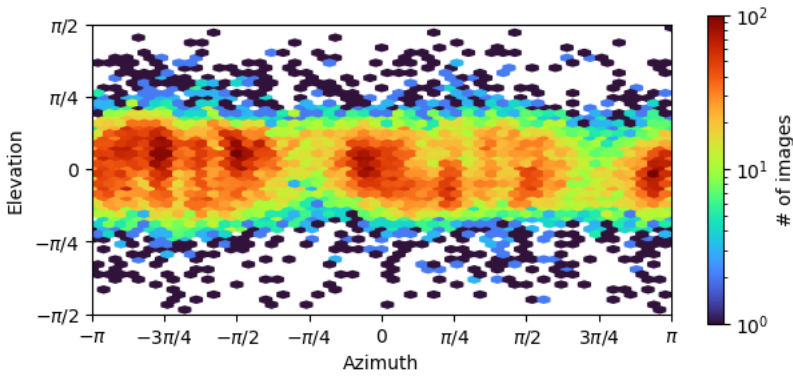
> [2026-06-09 15:23:58.49 GMT-4] Guinier Plot Iteration 002



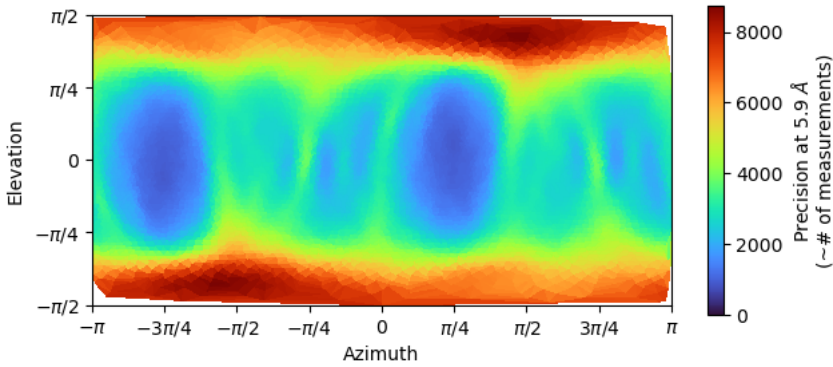
> [2026-06-09 15:23:58.91 GMT-4] Noise Model Iteration 002



> [2026-06-09 15:24:00.02 GMT-4] Viewing Direction Distribution Iteration 002

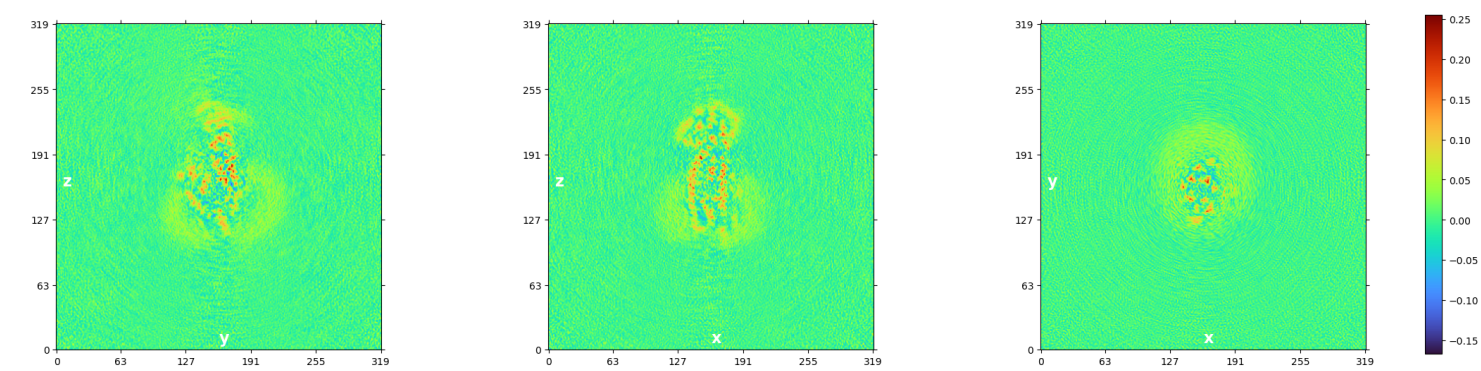


> [2026-06-09 15:24:01.72 GMT-4] Posterior Precision Directional Distribution Iteration 002

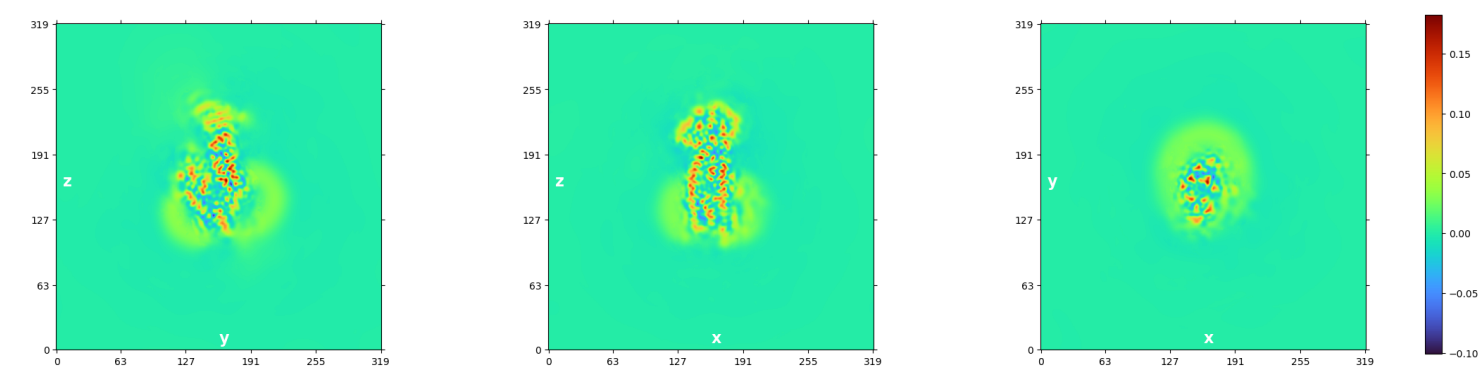


```
> [2026-06-09 15:24:01.72 GMT-4] [CPU: 10.93 GB] [Avail: 294.73 GB] Done in 20.523s.
> [2026-06-09 15:24:01.73 GMT-4] [CPU: 10.93 GB] [Avail: 294.73 GB] Outputting files..
> [2026-06-09 15:24:18.29 GMT-4] [CPU: 10.81 GB] [Avail: 291.42 GB] Done in 16.558s.
> [2026-06-09 15:24:18.29 GMT-4] [CPU: 10.81 GB] [Avail: 291.42 GB] Done iteration 2 in 202.248s. Total time so far 591.785s
> [2026-06-09 15:24:19.03 GMT-4] [CPU: 10.81 GB] [Avail: 291.59 GB] ----- Start Iteration 3
> [2026-06-09 15:24:19.03 GMT-4] [CPU: 10.81 GB] [Avail: 291.59 GB] Using Max Alignment Radius 72.190 (3.679A)
> [2026-06-09 15:24:19.03 GMT-4] [CPU: 10.81 GB] [Avail: 291.59 GB] Auto batchsize: 13930 in each split
> [2026-06-09 15:24:19.04 GMT-4] [CPU: 10.81 GB] [Avail: 291.58 GB] Using dynamic mask.
> [2026-06-09 15:24:49.83 GMT-4] [CPU: 10.81 GB] [Avail: 294.63 GB] -- THR 1 BATCH 500 NUM 3414 TOTAL 5.4652967 ELAPSED 55.295569
--
> [2026-06-09 15:25:47.36 GMT-4] [CPU: 11.57 GB] [Avail: 292.46 GB] Processed 27758.000 images in 57.527s.
> [2026-06-09 15:25:50.87 GMT-4] [CPU: 11.60 GB] [Avail: 293.18 GB] Computing FSCs...
> [2026-06-09 15:25:50.87 GMT-4] [CPU: 11.60 GB] [Avail: 293.18 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-09 15:25:50.87 GMT-4] [CPU: 11.60 GB] [Avail: 293.18 GB] Computing FFTs on GPU.
> [2026-06-09 15:25:52.87 GMT-4] [CPU: 11.60 GB] [Avail: 292.94 GB] Done in 2.000s
> [2026-06-09 15:25:52.87 GMT-4] [CPU: 11.60 GB] [Avail: 292.94 GB] Computing cFSCs...
> [2026-06-09 15:25:54.92 GMT-4] [CPU: 11.60 GB] [Avail: 292.93 GB] Done in 2.045s
> [2026-06-09 15:25:54.93 GMT-4] [CPU: 11.60 GB] [Avail: 292.93 GB] Using Filter Radius 79.431 (3.344A) | Previous: 72.190
(3.679A)
> [2026-06-09 15:26:01.48 GMT-4] [CPU: 10.84 GB] [Avail: 293.13 GB] Non-uniform regularization with compute option: GPU
> [2026-06-09 15:26:01.49 GMT-4] [CPU: 10.84 GB] [Avail: 293.13 GB] Running local cross validation for A ...
> [2026-06-09 15:26:24.25 GMT-4] [CPU: 11.06 GB] [Avail: 293.83 GB] Local cross validation A done in 22.758s
```


> [2026-06-09 15:26:25.71 GMT-4] FSC Filtered Side A



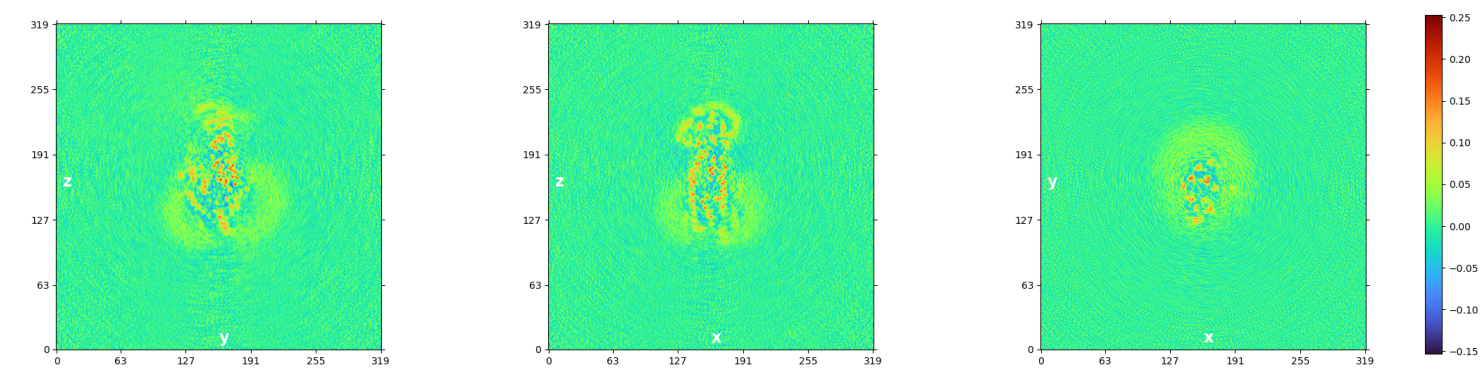
> [2026-06-09 15:26:26.82 GMT-4] CV Filtered Side A



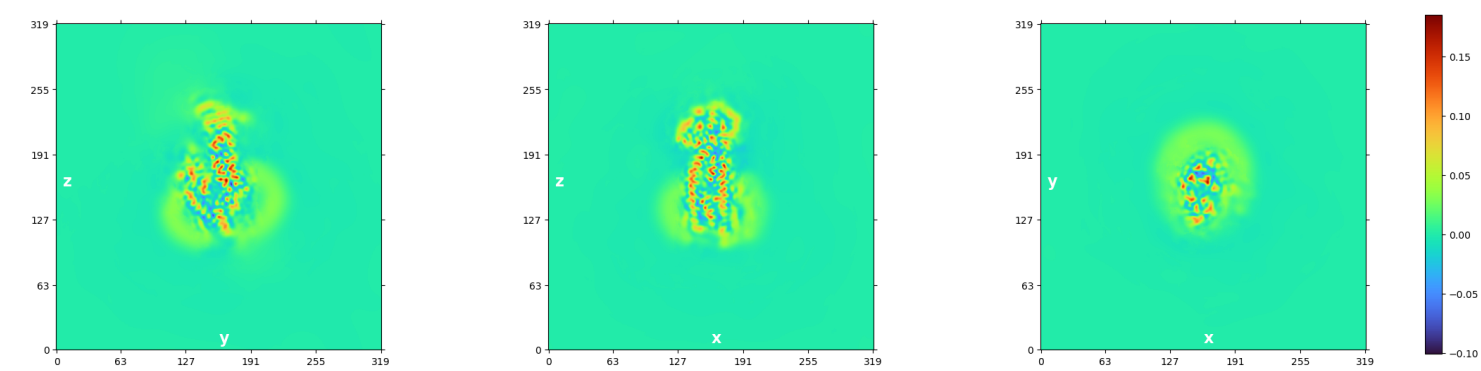
> [2026-06-09 15:26:26.83 GMT-4] [CPU: 11.06 GB] [Avail: 292.78 GB] Running local cross validation for B ...

> [2026-06-09 15:26:49.33 GMT-4] [CPU: 11.31 GB] [Avail: 291.94 GB] Local cross validation B done in 22.502s

> [2026-06-09 15:26:50.97 GMT-4] FSC Filtered Side B



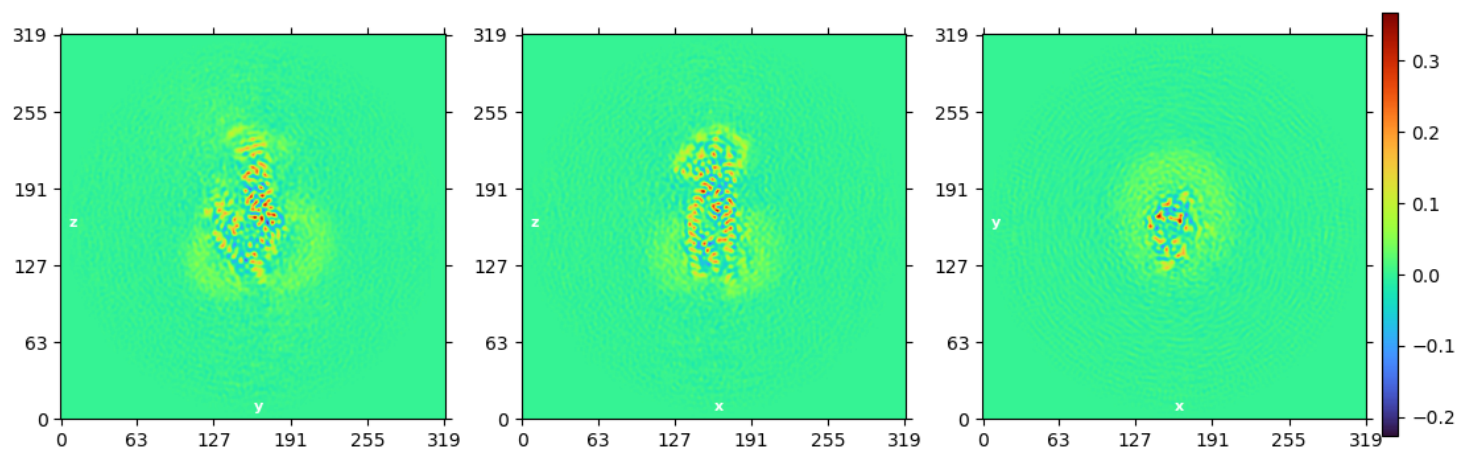
> [2026-06-09 15:26:52.05 GMT-4] CV Filtered Side B



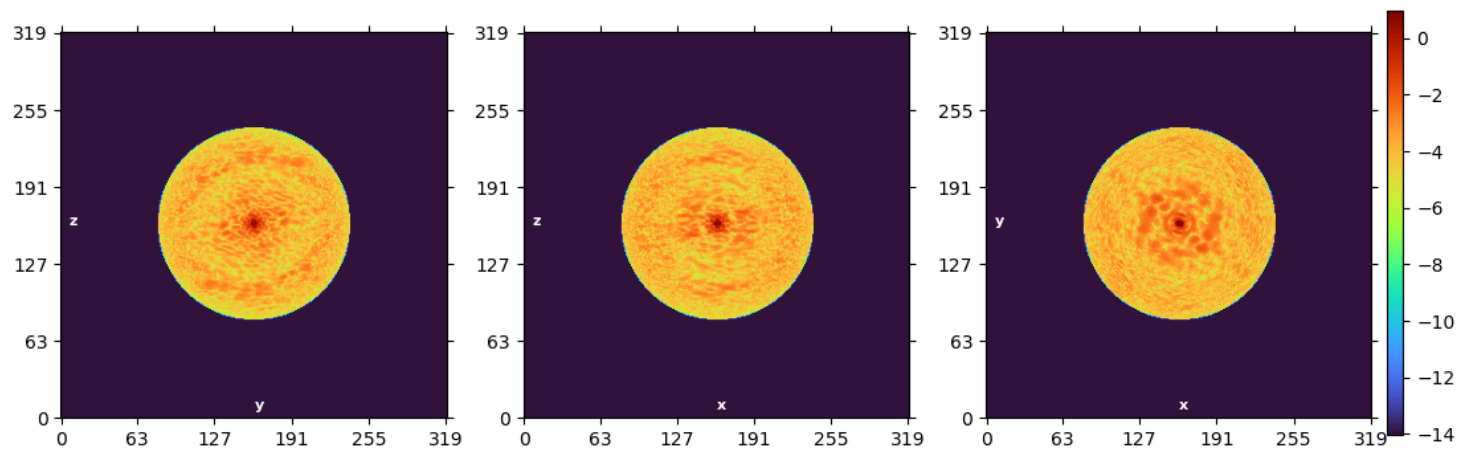
> [2026-06-09 15:27:03.02 GMT-4] [CPU: 11.31 GB] [Avail: 292.18 GB] Estimated Bfactor: -85.7

> [2026-06-09 15:27:03.03 GMT-4] [CPU: 11.31 GB] [Avail: 292.18 GB] Plotting..

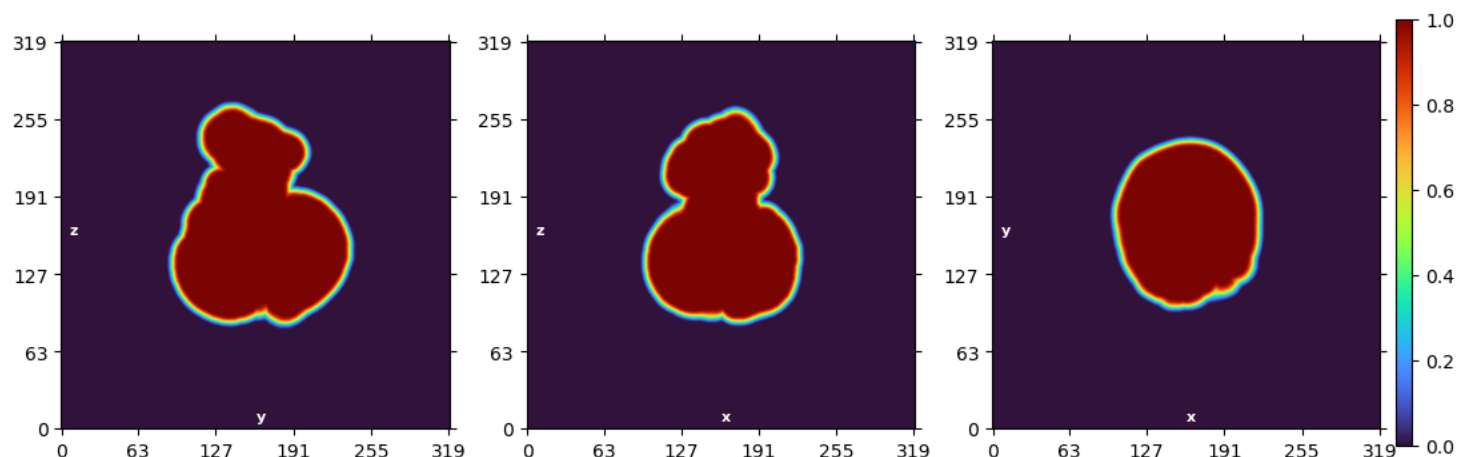
> [2026-06-09 15:27:10.22 GMT-4] Real Space Slices Iteration 003



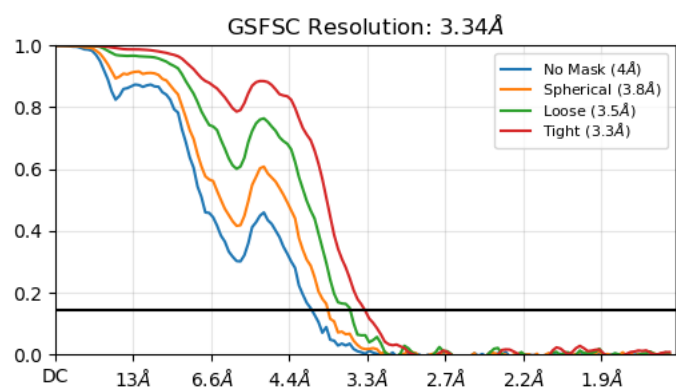
> [2026-06-09 15:27:11.72 GMT-4] Fourier Space Slices Iteration 003



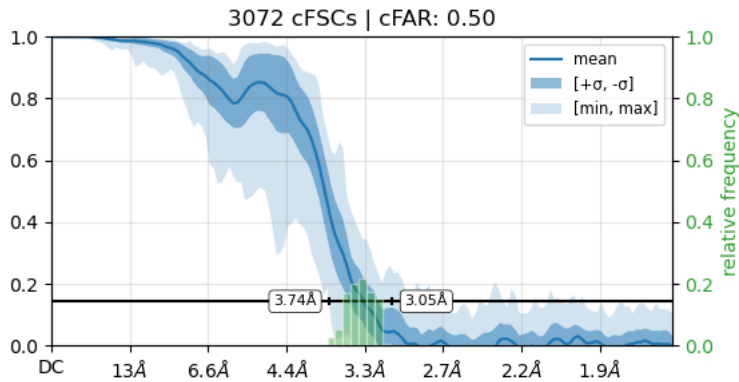
> [2026-06-09 15:27:14.12 GMT-4] Real Space Mask Slices Iteration 003



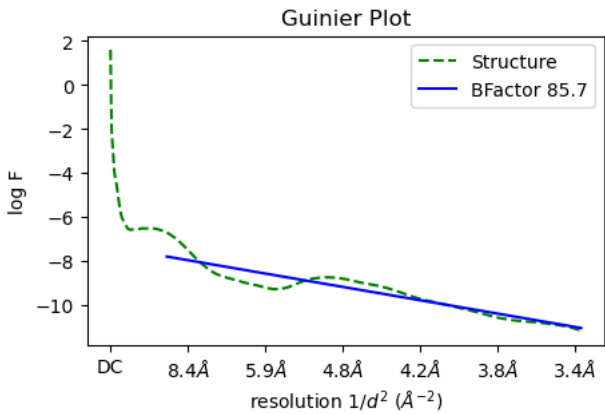
> [2026-06-09 15:27:14.67 GMT-4] FSC Iteration 003



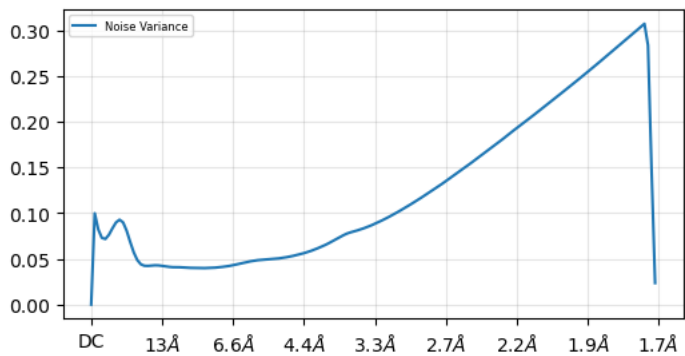
> [2026-06-09 15:27:15.66 GMT-4] cFSCs (Half-angle: 20°) Iteration 003, with tight mask



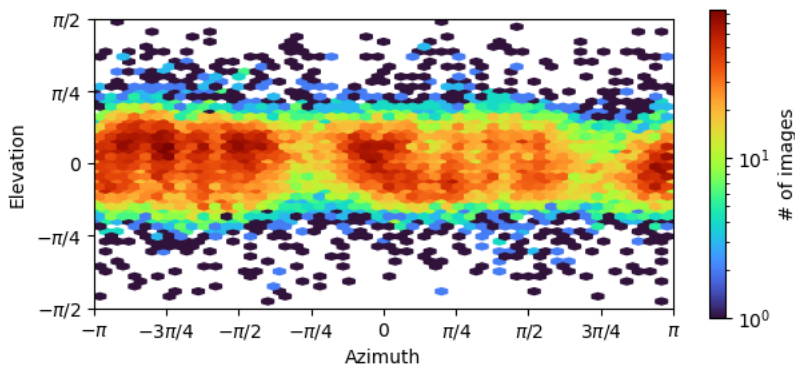
> [2026-06-09 15:27:21.71 GMT-4] Guinier Plot Iteration 003



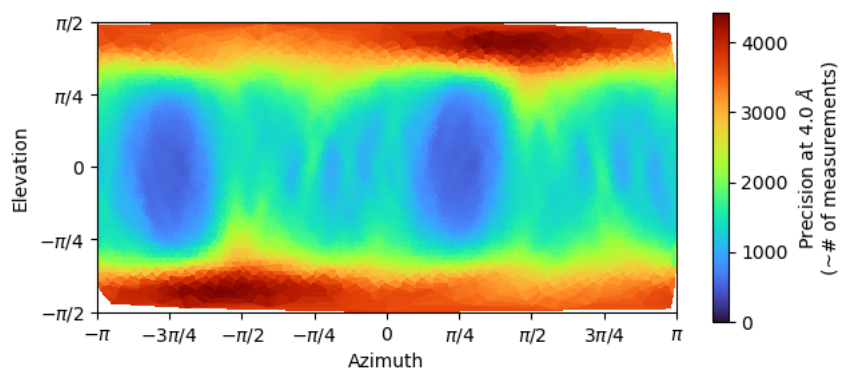
> [2026-06-09 15:27:22.08 GMT-4] Noise Model Iteration 003



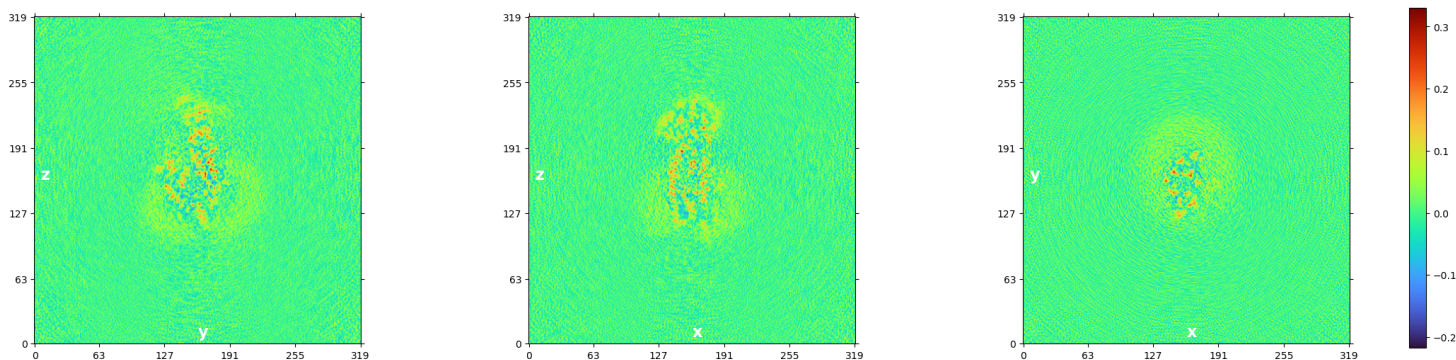
> [2026-06-09 15:27:23.06 GMT-4] Viewing Direction Distribution Iteration 003



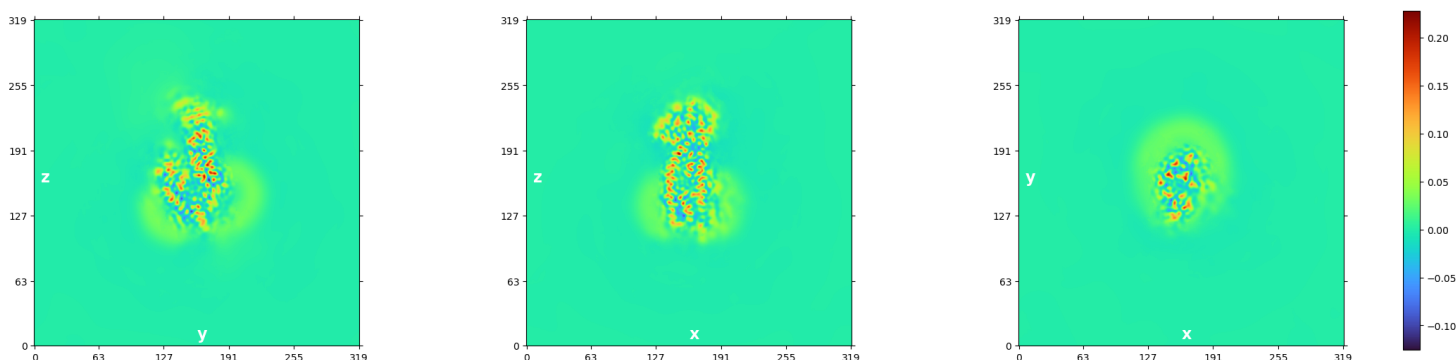
> [2026-06-09 15:27:24.72 GMT-4] Posterior Precision Directional Distribution Iteration 003



```
> [2026-06-09 15:27:24.73 GMT-4] [CPU: 11.34 GB] [Avail: 290.63 GB] Done in 21.692s.
> [2026-06-09 15:27:24.73 GMT-4] [CPU: 11.34 GB] [Avail: 290.63 GB] Outputting files..
> [2026-06-09 15:27:41.02 GMT-4] [CPU: 11.25 GB] [Avail: 290.64 GB] Done in 16.283s.
> [2026-06-09 15:27:41.02 GMT-4] [CPU: 11.25 GB] [Avail: 290.64 GB] Done iteration 3 in 201.992s. Total time so far 794.513s
> [2026-06-09 15:27:41.82 GMT-4] [CPU: 11.25 GB] [Avail: 289.83 GB] ----- Start Iteration 4
> [2026-06-09 15:27:41.82 GMT-4] [CPU: 11.25 GB] [Avail: 289.82 GB] Using Max Alignment Radius 79.431 (3.344A)
> [2026-06-09 15:27:41.83 GMT-4] [CPU: 11.25 GB] [Avail: 289.81 GB] Auto batchsize: 13930 in each split
> [2026-06-09 15:27:41.83 GMT-4] [CPU: 11.25 GB] [Avail: 289.81 GB] Using dynamic mask.
> [2026-06-09 15:28:14.53 GMT-4] [CPU: 11.25 GB] [Avail: 291.46 GB] -- THR 1 BATCH 500 NUM 3500 TOTAL 8.0075228 ELAPSED 56.765657
--
> [2026-06-09 15:29:13.12 GMT-4] [CPU: 12.00 GB] [Avail: 303.88 GB] Processed 27758.000 images in 58.590s.
> [2026-06-09 15:29:16.31 GMT-4] [CPU: 12.03 GB] [Avail: 302.93 GB] Computing FSCs...
> [2026-06-09 15:29:16.31 GMT-4] [CPU: 12.03 GB] [Avail: 302.92 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-09 15:29:16.32 GMT-4] [CPU: 12.03 GB] [Avail: 302.92 GB] Computing FFTs on GPU.
> [2026-06-09 15:29:18.25 GMT-4] [CPU: 12.03 GB] [Avail: 301.87 GB] Done in 1.932s
> [2026-06-09 15:29:18.25 GMT-4] [CPU: 12.03 GB] [Avail: 301.87 GB] Computing cFSCs...
> [2026-06-09 15:29:20.38 GMT-4] [CPU: 12.04 GB] [Avail: 299.97 GB] Done in 2.129s
> [2026-06-09 15:29:20.39 GMT-4] [CPU: 12.04 GB] [Avail: 299.97 GB] Using Filter Radius 82.409 (3.223A) | Previous: 79.431
(3.344A)
> [2026-06-09 15:29:27.37 GMT-4] [CPU: 11.28 GB] [Avail: 301.89 GB] Non-uniform regularization with compute option: GPU
> [2026-06-09 15:29:27.37 GMT-4] [CPU: 11.28 GB] [Avail: 301.88 GB] Running local cross validation for A ...
> [2026-06-09 15:29:50.16 GMT-4] [CPU: 11.50 GB] [Avail: 299.20 GB] Local cross validation A done in 22.782s
> [2026-06-09 15:29:51.60 GMT-4] FSC Filtered Side A
```



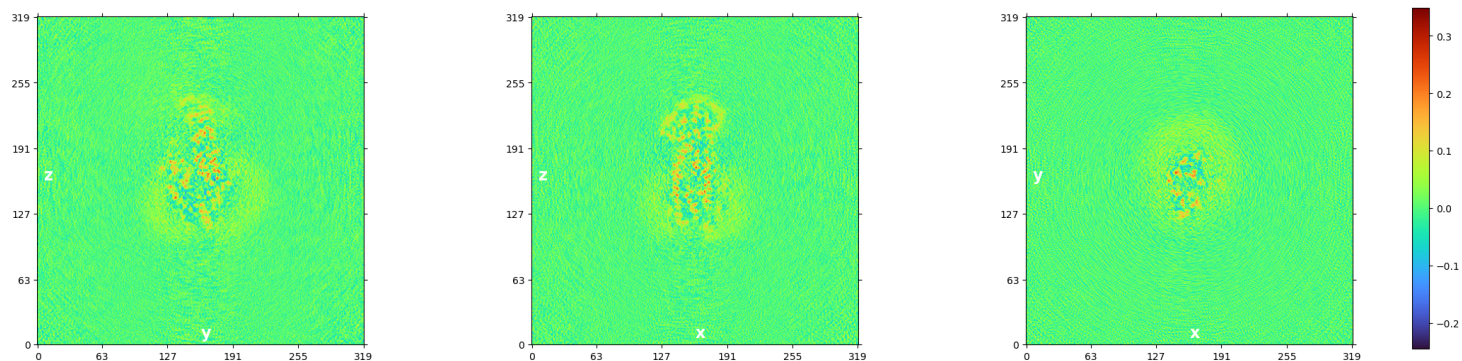
> [2026-06-09 15:29:52.78 GMT-4] CV Filtered Side A



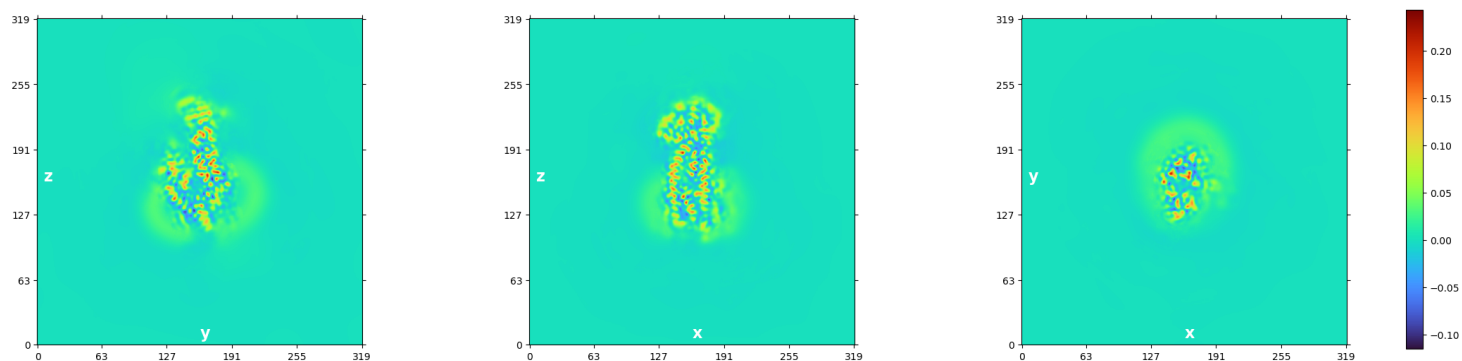
> [2026-06-09 15:29:52.78 GMT-4] [CPU: 11.50 GB] [Avail: 298.90 GB] Running local cross validation for B ...

> [2026-06-09 15:30:16.17 GMT-4] [CPU: 11.75 GB] [Avail: 289.00 GB] Local cross validation B done in 23.380s

> [2026-06-09 15:30:17.88 GMT-4] FSC Filtered Side B



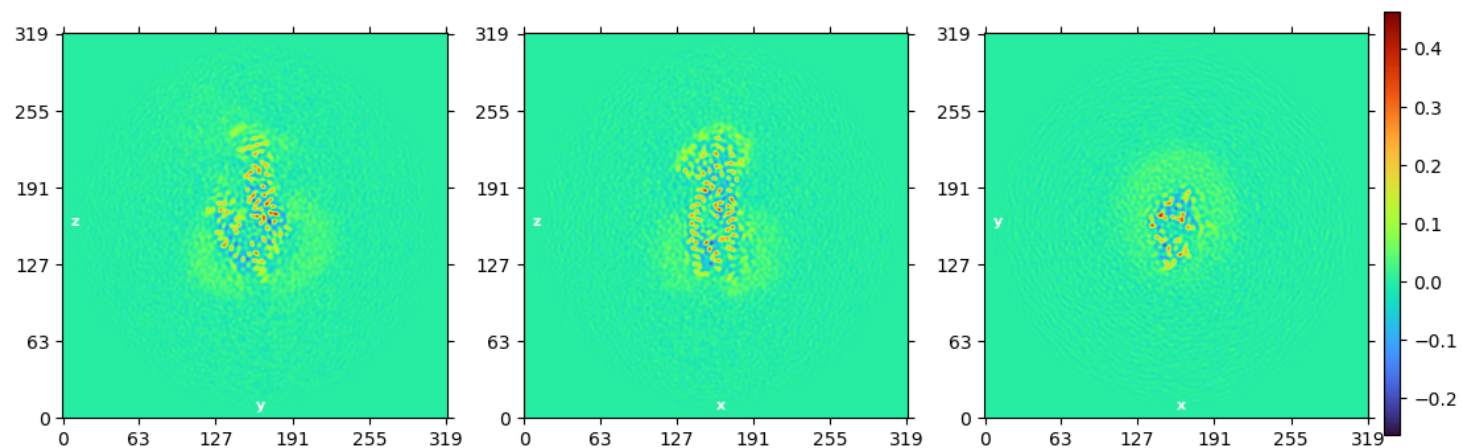
> [2026-06-09 15:30:19.09 GMT-4] CV Filtered Side B



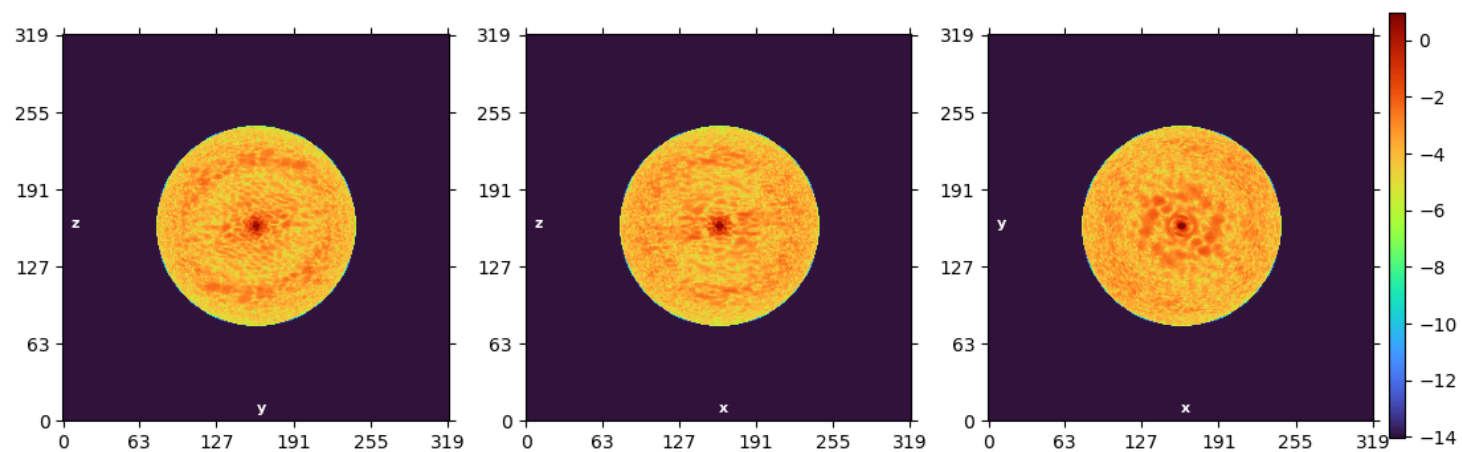
> [2026-06-09 15:30:30.77 GMT-4] [CPU: 11.75 GB] [Avail: 291.50 GB] Estimated Bfactor: -75.0

> [2026-06-09 15:30:30.77 GMT-4] [CPU: 11.75 GB] [Avail: 291.50 GB] Plotting..

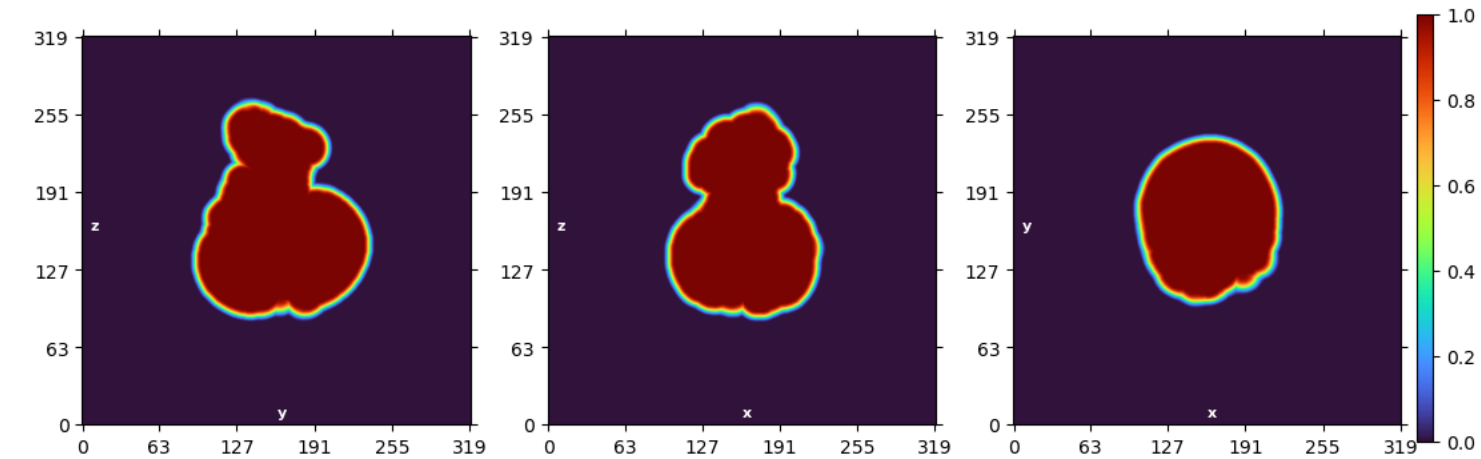
> [2026-06-09 15:30:37.47 GMT-4] Real Space Slices Iteration 004



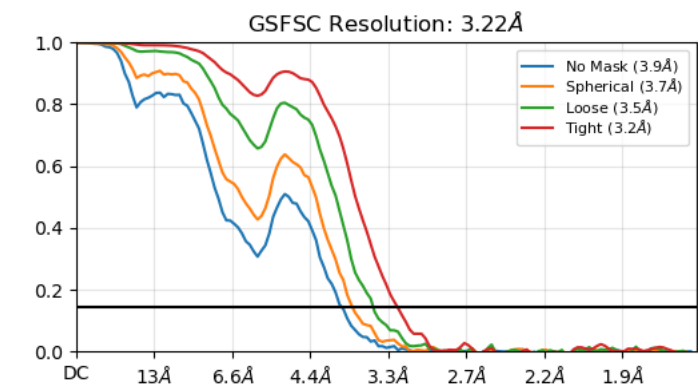
> [2026-06-09 15:30:38.85 GMT-4] Fourier Space Slices Iteration 004



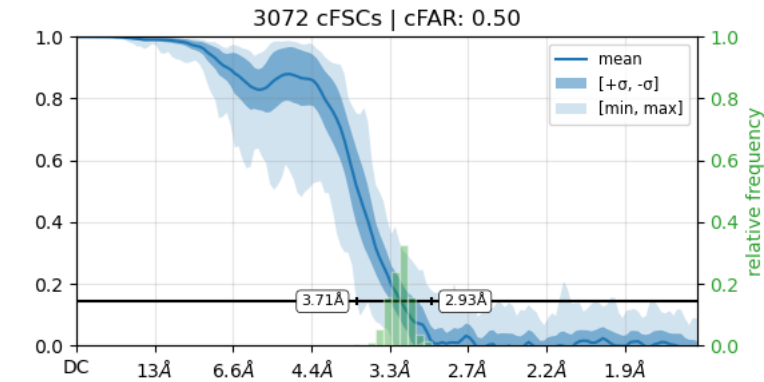
> [2026-06-09 15:30:40.87 GMT-4] Real Space Mask Slices Iteration 004



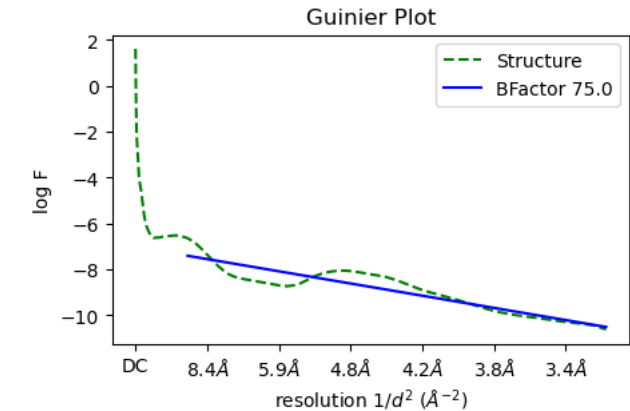
> [2026-06-09 15:30:41.46 GMT-4] FSC Iteration 004



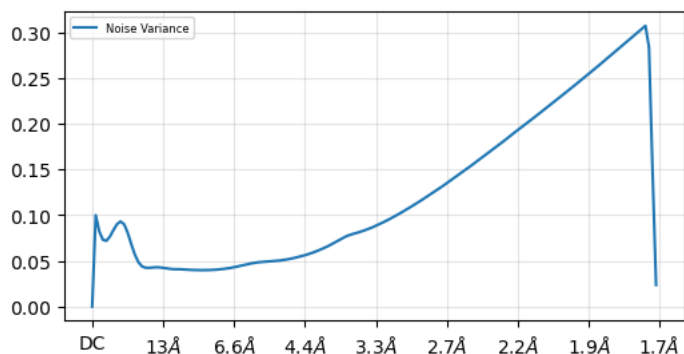
> [2026-06-09 15:30:42.29 GMT-4] cFSCs (Half-angle: 20°) Iteration 004, with tight mask



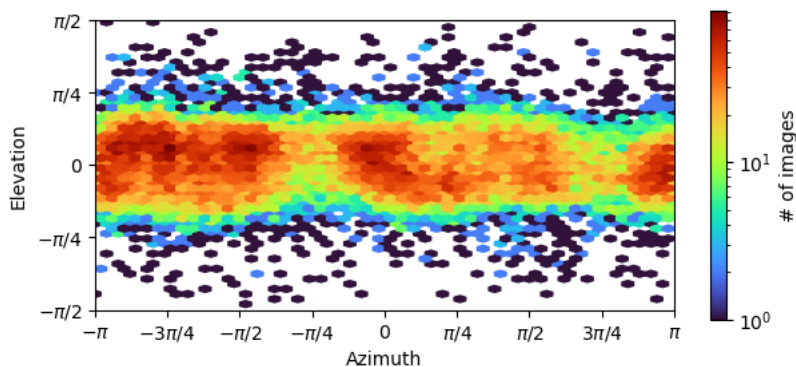
> [2026-06-09 15:30:48.88 GMT-4] Guinier Plot Iteration 004



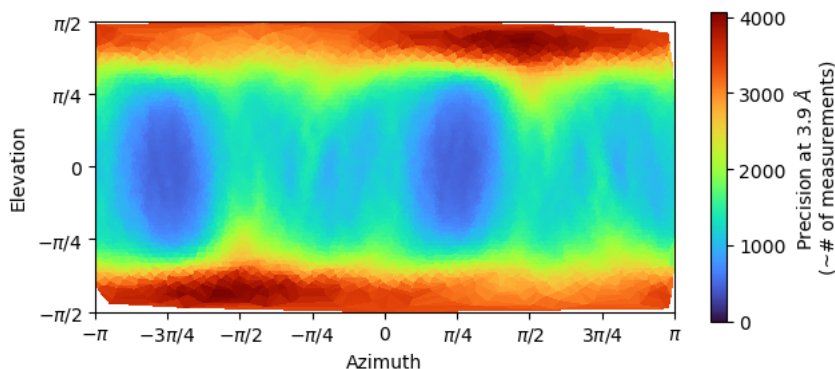
> [2026-06-09 15:30:49.31 GMT-4] Noise Model Iteration 004



> [2026-06-09 15:30:50.33 GMT-4] Viewing Direction Distribution Iteration 004



> [2026-06-09 15:30:52.15 GMT-4] Posterior Precision Directional Distribution Iteration 004

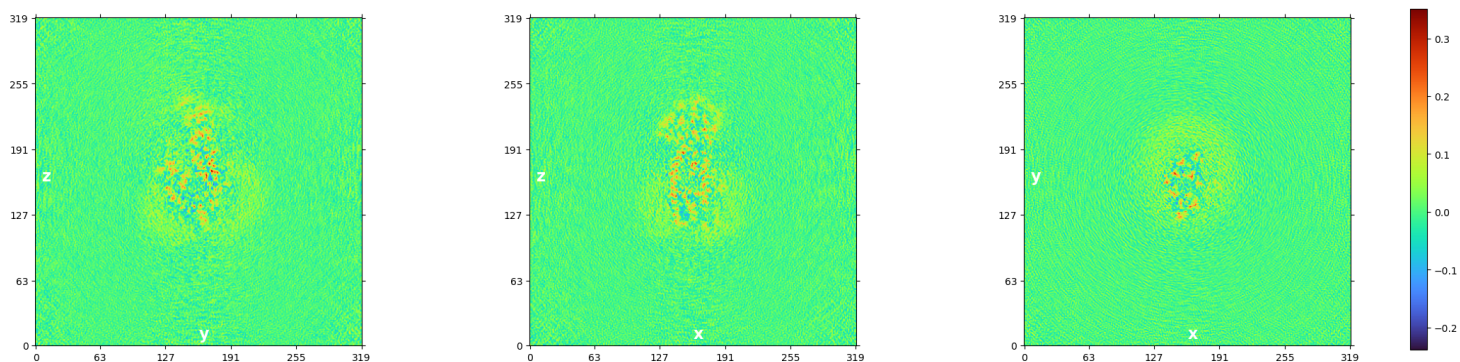


```
> [2026-06-09 15:30:52.15 GMT-4] [CPU: 11.78 GB] [Avail: 298.05 GB] Done in 21.376s.
> [2026-06-09 15:30:52.16 GMT-4] [CPU: 11.78 GB] [Avail: 298.05 GB] Outputting files..
> [2026-06-09 15:31:08.74 GMT-4] [CPU: 11.69 GB] [Avail: 300.68 GB] Done in 16.581s.
> [2026-06-09 15:31:08.74 GMT-4] [CPU: 11.69 GB] [Avail: 300.68 GB] Done iteration 4 in 206.919s. Total time so far 1002.235s
> [2026-06-09 15:31:09.64 GMT-4] [CPU: 11.69 GB] [Avail: 298.82 GB] ----- Start Iteration 5
> [2026-06-09 15:31:09.64 GMT-4] [CPU: 11.69 GB] [Avail: 298.81 GB] Using Max Alignment Radius 82.409 (3.223A)
> [2026-06-09 15:31:09.65 GMT-4] [CPU: 11.69 GB] [Avail: 298.79 GB] Using Full Dataset (split 13930 in A, 13828 in B)
> [2026-06-09 15:31:09.65 GMT-4] [CPU: 11.69 GB] [Avail: 298.79 GB] Using dynamic mask.
> [2026-06-09 15:31:41.75 GMT-4] [CPU: 11.69 GB] [Avail: 298.20 GB] -- THR 1 BATCH 500 NUM 3414 TOTAL 7.8533134 ELAPSED 56.514362
--
> [2026-06-09 15:32:40.67 GMT-4] [CPU: 12.44 GB] [Avail: 289.91 GB] Processed 27758.000 images in 58.929s.
> [2026-06-09 15:32:44.15 GMT-4] [CPU: 12.47 GB] [Avail: 289.71 GB] Computing FSCs...
> [2026-06-09 15:32:44.15 GMT-4] [CPU: 12.47 GB] [Avail: 289.71 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-09 15:32:44.16 GMT-4] [CPU: 12.47 GB] [Avail: 289.71 GB] Computing FFTs on GPU.
> [2026-06-09 15:32:47.49 GMT-4] [CPU: 12.47 GB] [Avail: 289.15 GB] Done in 3.334s
> [2026-06-09 15:32:47.51 GMT-4] [CPU: 12.47 GB] [Avail: 289.12 GB] Optimizing FSC Mask...
> [2026-06-09 15:32:47.51 GMT-4] [CPU: 12.47 GB] [Avail: 289.11 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-09 15:32:47.52 GMT-4] [CPU: 12.47 GB] [Avail: 289.11 GB] Computing FFTs on GPU.
> [2026-06-09 15:32:49.50 GMT-4] [CPU: 12.72 GB] [Avail: 287.11 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-09 15:32:49.51 GMT-4] [CPU: 12.72 GB] [Avail: 287.11 GB] Computing FFTs on GPU.
> [2026-06-09 15:32:51.55 GMT-4] [CPU: 12.72 GB] [Avail: 287.41 GB] Done in 4.033s
```

```

> [2026-06-09 15:32:51.55 GMT-4] [CPU: 12.72 GB] [Avail: 287.41 GB] Computing cFSCs...
> [2026-06-09 15:32:53.60 GMT-4] [CPU: 12.72 GB] [Avail: 287.20 GB] Done in 2.049s
> [2026-06-09 15:32:53.61 GMT-4] [CPU: 12.72 GB] [Avail: 287.20 GB] Using Filter Radius 85.620 (3.102A) | Previous: 82.409
(3.223A)
> [2026-06-09 15:32:59.84 GMT-4] [CPU: 11.97 GB] [Avail: 286.61 GB] Non-uniform regularization with compute option: GPU
> [2026-06-09 15:32:59.84 GMT-4] [CPU: 11.97 GB] [Avail: 286.61 GB] Running local cross validation for A ...
> [2026-06-09 15:33:23.08 GMT-4] [CPU: 12.19 GB] [Avail: 286.68 GB] Local cross validation A done in 23.230s
> [2026-06-09 15:33:24.63 GMT-4] FSC Filtered Side A

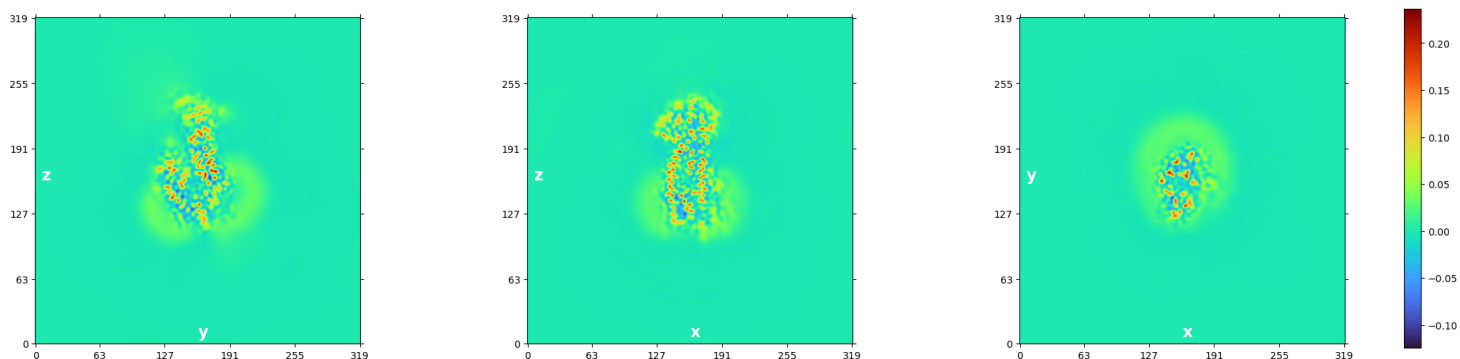
```



```

> [2026-06-09 15:33:25.68 GMT-4] CV Filtered Side A

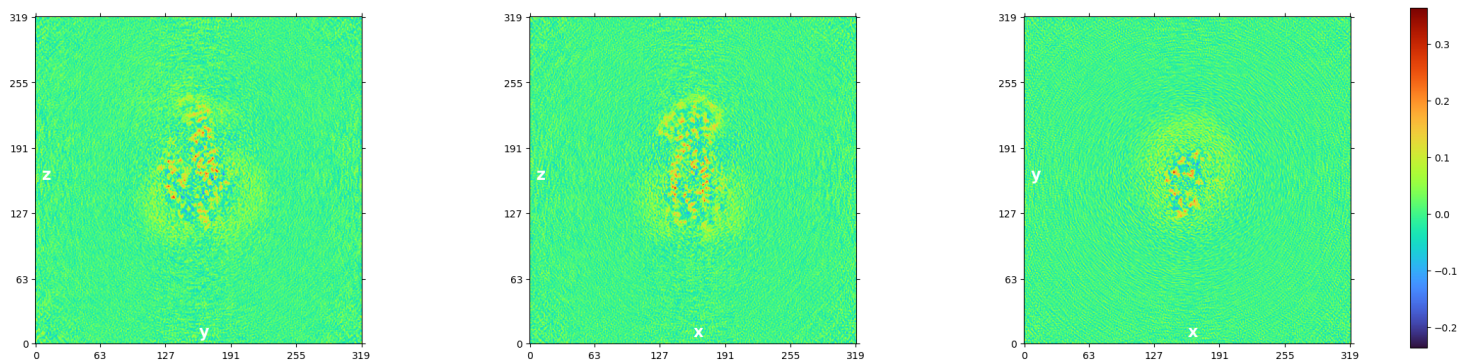
```



```

> [2026-06-09 15:33:25.68 GMT-4] [CPU: 12.19 GB] [Avail: 286.25 GB] Running local cross validation for B ...
> [2026-06-09 15:33:47.94 GMT-4] [CPU: 12.44 GB] [Avail: 287.44 GB] Local cross validation B done in 22.249s
> [2026-06-09 15:33:49.56 GMT-4] FSC Filtered Side B

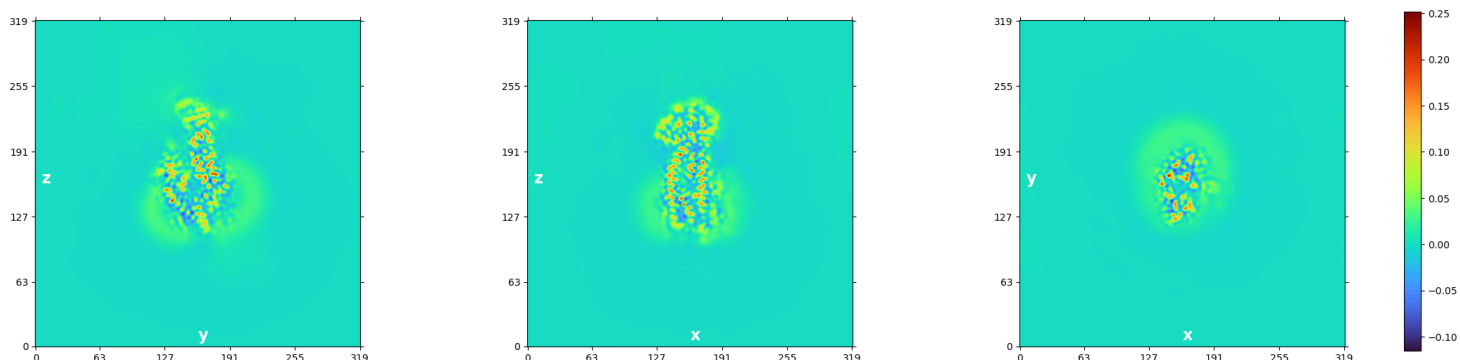
```



```

> [2026-06-09 15:33:50.85 GMT-4] CV Filtered Side B

```

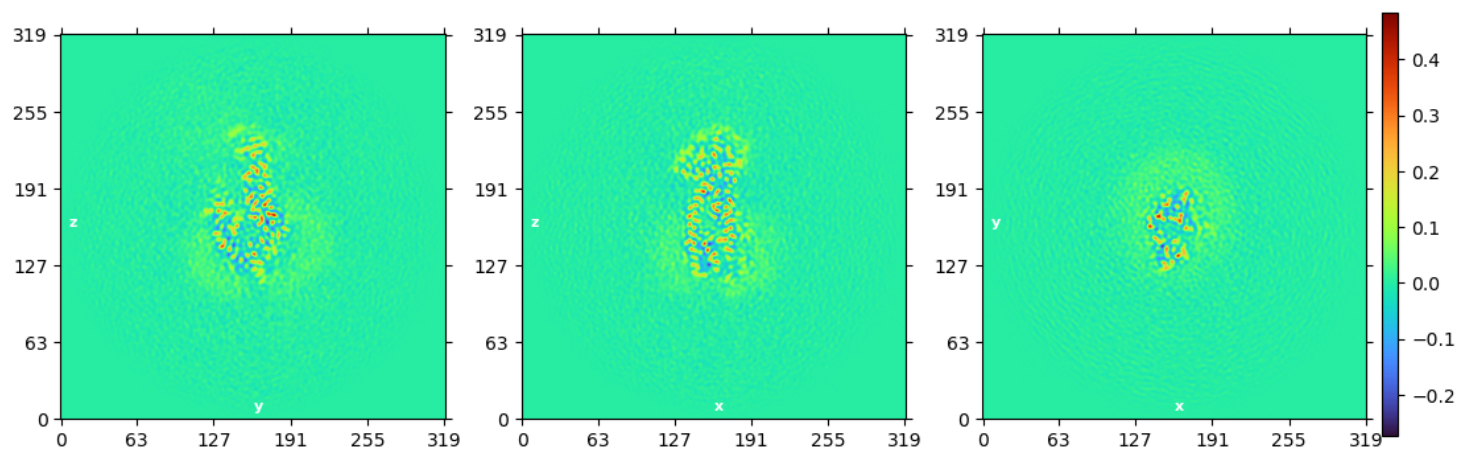


```

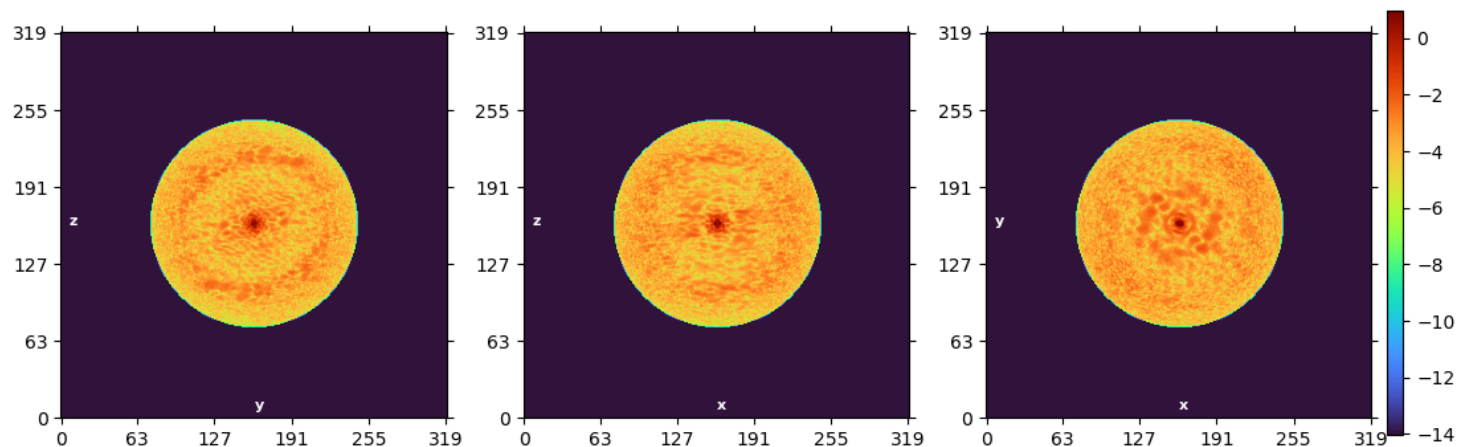
> [2026-06-09 15:34:02.17 GMT-4] [CPU: 12.44 GB] [Avail: 286.91 GB] Estimated Bfactor: -71.3
> [2026-06-09 15:34:02.18 GMT-4] [CPU: 12.44 GB] [Avail: 286.89 GB] Plotting..

```

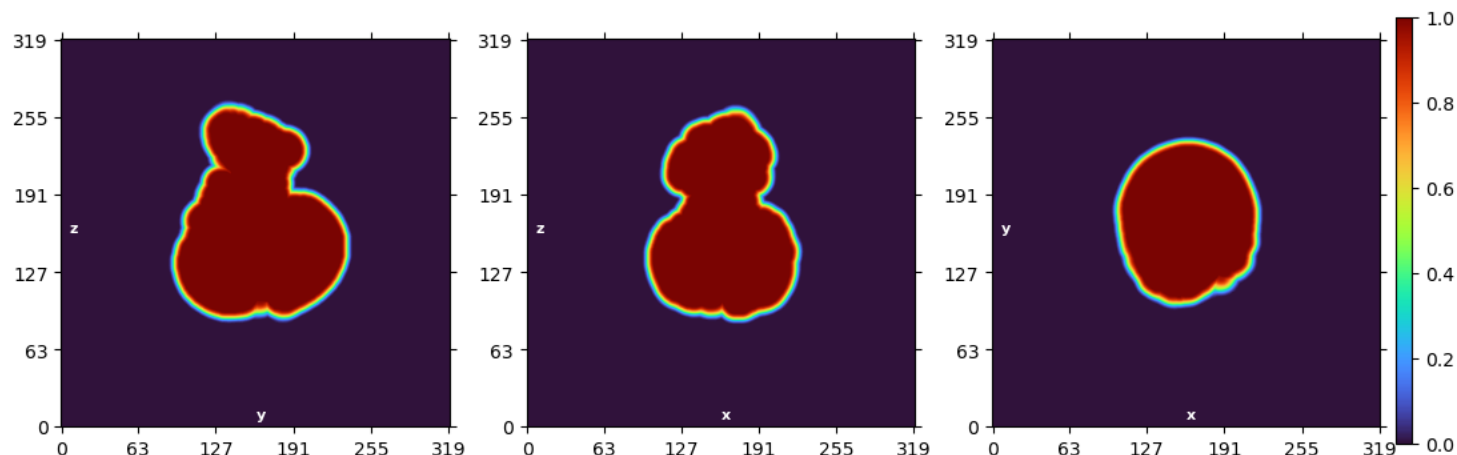

> [2026-06-09 15:34:09.03 GMT-4] Real Space Slices Iteration 005



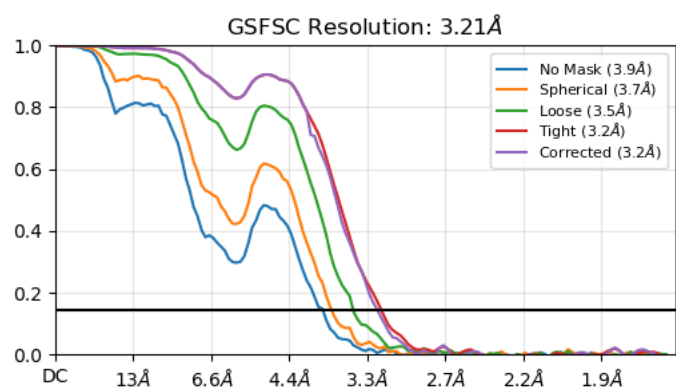
> [2026-06-09 15:34:10.53 GMT-4] Fourier Space Slices Iteration 005



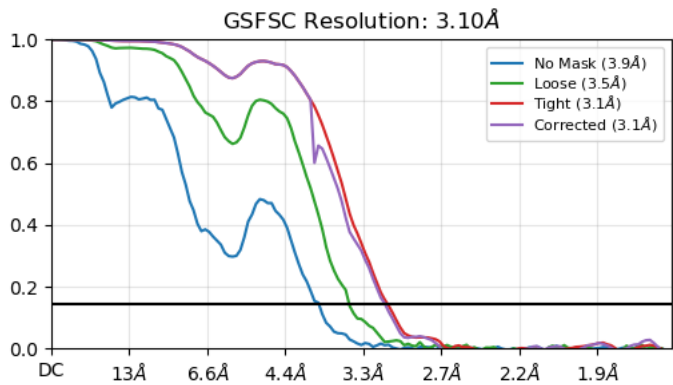
> [2026-06-09 15:34:12.72 GMT-4] Real Space Mask Slices Iteration 005



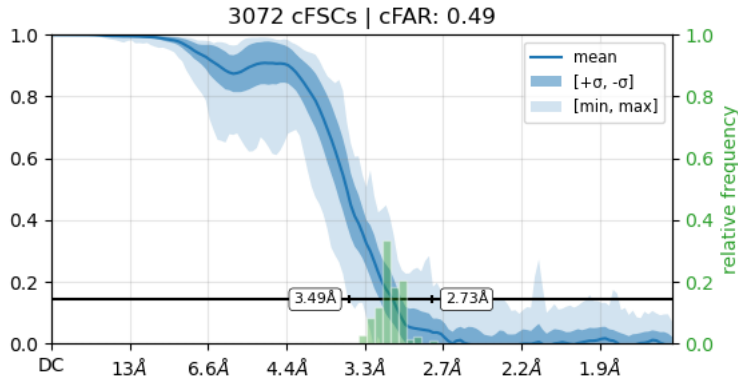
> [2026-06-09 15:34:13.26 GMT-4] FSC Iteration 005



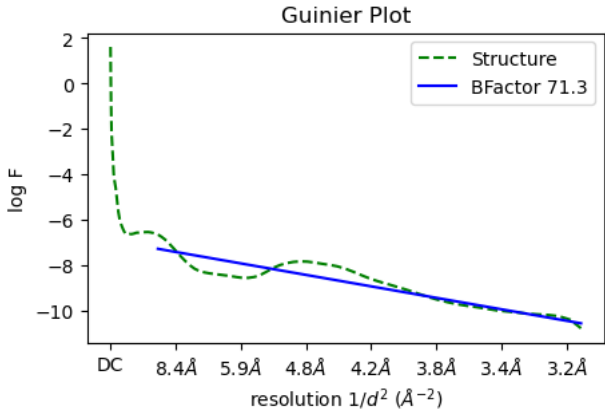
> [2026-06-09 15:34:13.86 GMT-4] FSC Iteration 005, after FSC-mask auto-tightening



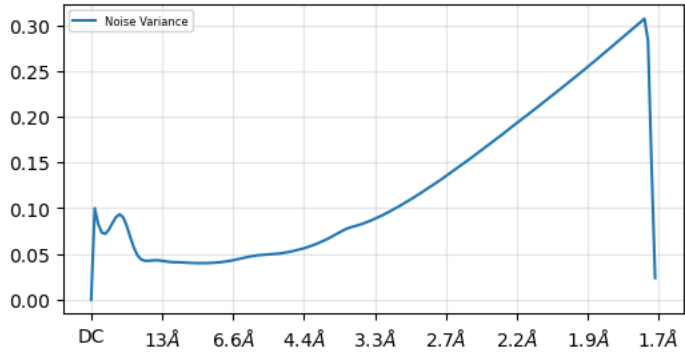
> [2026-06-09 15:34:14.82 GMT-4] cFSCs (Half-angle: 20°) Iteration 005, with autotight mask



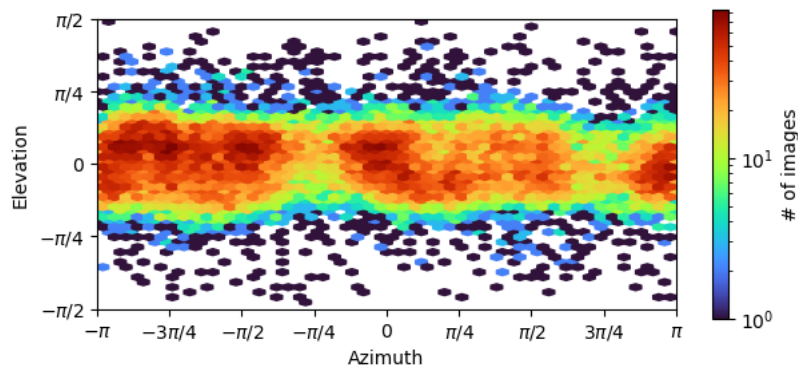
> [2026-06-09 15:34:21.06 GMT-4] Guinier Plot Iteration 005



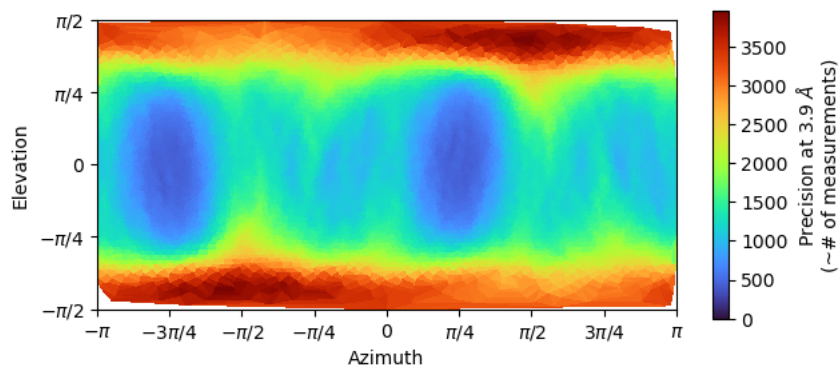
> [2026-06-09 15:34:21.49 GMT-4] Noise Model Iteration 005



> [2026-06-09 15:34:22.57 GMT-4] Viewing Direction Distribution Iteration 005



> [2026-06-09 15:34:24.29 GMT-4] Posterior Precision Directional Distribution Iteration 005



```
> [2026-06-09 15:34:24.29 GMT-4] [CPU: 12.47 GB] [Avail: 284.72 GB] Done in 22.105s.
> [2026-06-09 15:34:24.29 GMT-4] [CPU: 12.47 GB] [Avail: 284.71 GB] Outputting files..
> [2026-06-09 15:34:42.43 GMT-4] [CPU: 12.38 GB] [Avail: 286.65 GB] Done in 18.131s.
> [2026-06-09 15:34:42.43 GMT-4] [CPU: 12.38 GB] [Avail: 286.69 GB] Done iteration 5 in 212.787s. Total time so far 1215.922s
> [2026-06-09 15:34:43.21 GMT-4] [CPU: 12.38 GB] [Avail: 286.78 GB] ===== Done Refinement =====
> [2026-06-09 15:34:43.22 GMT-4] [CPU: 12.38 GB] [Avail: 286.78 GB] Note that the output structure from refinement has been
                                                                    lowpass filtered to the gold-standard FSC resolution
                                                                    estimated
                                                                    above. However, the structure probably needs sharpening in
                                                                    order to best visualize high resolution features. This can be
                                                                    done with the Sharpening task (as a new experiment).

> [2026-06-09 15:34:43.22 GMT-4] [CPU: 12.38 GB] [Avail: 286.77 GB] Full run took 1216.713s
> [2026-06-09 15:34:43.84 GMT-4] [CPU: 6.52 GB] [Avail: 292.52 GB] -----
-

> [2026-06-09 15:34:43.85 GMT-4] [CPU: 6.52 GB] [Avail: 292.52 GB] Compiling job outputs...
> [2026-06-09 15:34:43.85 GMT-4] [CPU: 6.52 GB] [Avail: 292.52 GB] Passing through outputs for output group particles from input
                                                                    group particles

> [2026-06-09 15:34:43.95 GMT-4] [CPU: 6.52 GB] [Avail: 292.57 GB] This job outputted results ['alignments3D', 'ctf']
> [2026-06-09 15:34:43.95 GMT-4] [CPU: 6.52 GB] [Avail: 292.58 GB] Loaded output dataset with 27758 items
> [2026-06-09 15:34:43.96 GMT-4] [CPU: 6.52 GB] [Avail: 292.58 GB] Passthrough results ['blob', 'alignments3D_multi']
> [2026-06-09 15:34:44.13 GMT-4] [CPU: 6.52 GB] [Avail: 292.71 GB] Loaded passthrough dataset with 27758 items
> [2026-06-09 15:34:44.17 GMT-4] [CPU: 6.52 GB] [Avail: 292.81 GB] Intersection of output and passthrough has 27758 items
> [2026-06-09 15:34:44.24 GMT-4] [CPU: 6.52 GB] [Avail: 292.74 GB] Output dataset contains: ['alignments3D_multi', 'blob']
> [2026-06-09 15:34:44.25 GMT-4] [CPU: 6.52 GB] [Avail: 292.74 GB] Outputting passthrough result blob
> [2026-06-09 15:34:44.25 GMT-4] [CPU: 6.52 GB] [Avail: 292.73 GB] Outputting passthrough result alignments3D_multi
> [2026-06-09 15:34:44.26 GMT-4] [CPU: 6.52 GB] [Avail: 292.72 GB] Checking outputs for output group particles
> [2026-06-09 15:34:44.41 GMT-4] [CPU: 6.52 GB] [Avail: 292.59 GB] Updating job size...
> [2026-06-09 15:34:45.24 GMT-4] [CPU: 6.52 GB] [Avail: 292.69 GB] Exporting job and creating csg files...
> [2026-06-09 15:34:46.11 GMT-4] [CPU: 6.52 GB] [Avail: 292.67 GB] *****
                                                                    **

> [2026-06-09 15:34:46.11 GMT-4] [CPU: 6.52 GB] [Avail: 292.67 GB] Job complete. Total time 1227.06s
```